

Computer Networks



Course Code: CSC263
Year/ Semester: II/IV

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Credit Hours: 3hrs



► Unit - 04: Network Layer

Class Load : 10 hrs

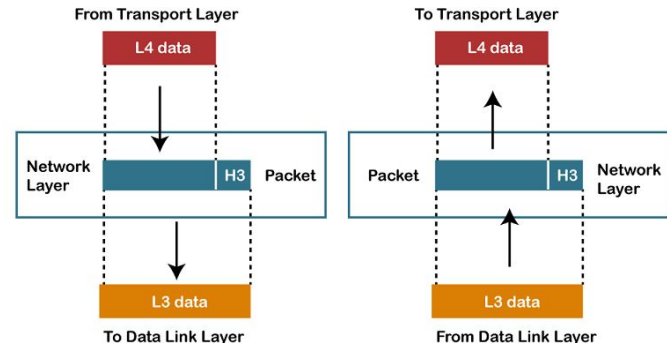
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► Network Layer:

- The transport layer passes the TCP segments or UDP datagram to the network layers.
- Then NL accept the transport layer **segments** and add **logical addressing** and form a **packet**.
- Responsible for **moving packets** from source to destination as it determines the best path to move data from source to the destination based on the **network conditions**, the **priority of service**, and **other factors**.
- It divides the outgoing messages into packets and assembles the incoming packets into messages for higher levels.

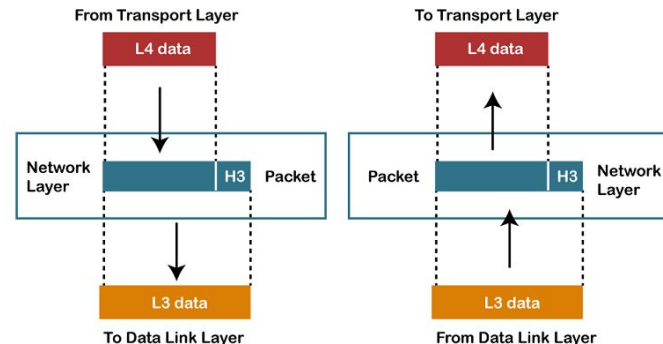
- **PDU : Packets**

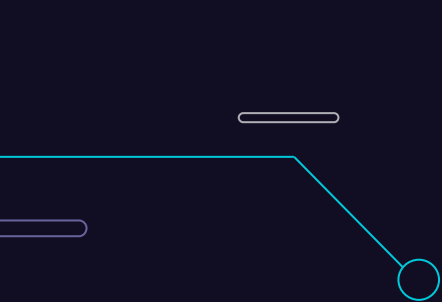


► Network Layer:

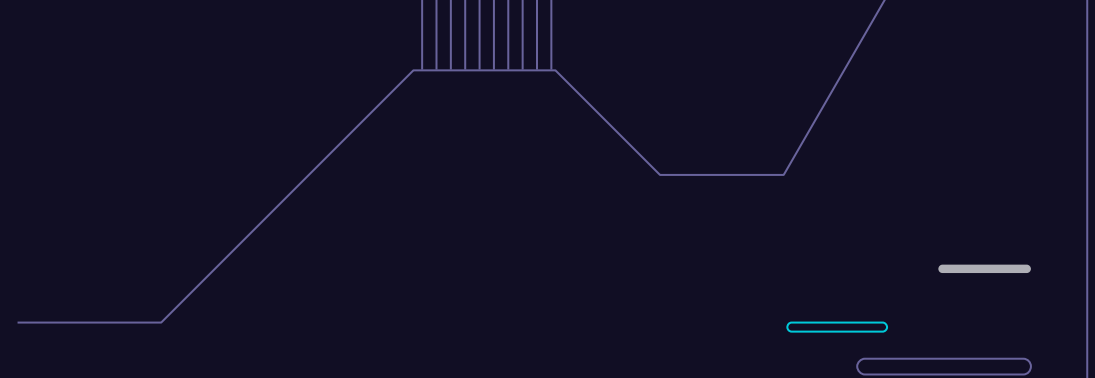
Functions of the Network Layer :

- **Routing:** The network layer protocols determine which route is suitable from source to destination. This function of the network layer is known as routing.
- **Logical Addressing:** To identify each device inter-network uniquely, the network layer defines an addressing scheme. The sender & receiver's IP addresses are placed in the header by the network layer. Such an address distinguishes each device uniquely and universally.
- **Packetizing:** A Network Layer receives the data from the upper layer and converts them into packets. This process is known as Packetizing. It is achieved by internet protocol (IP).





▶ Introduction & Functions



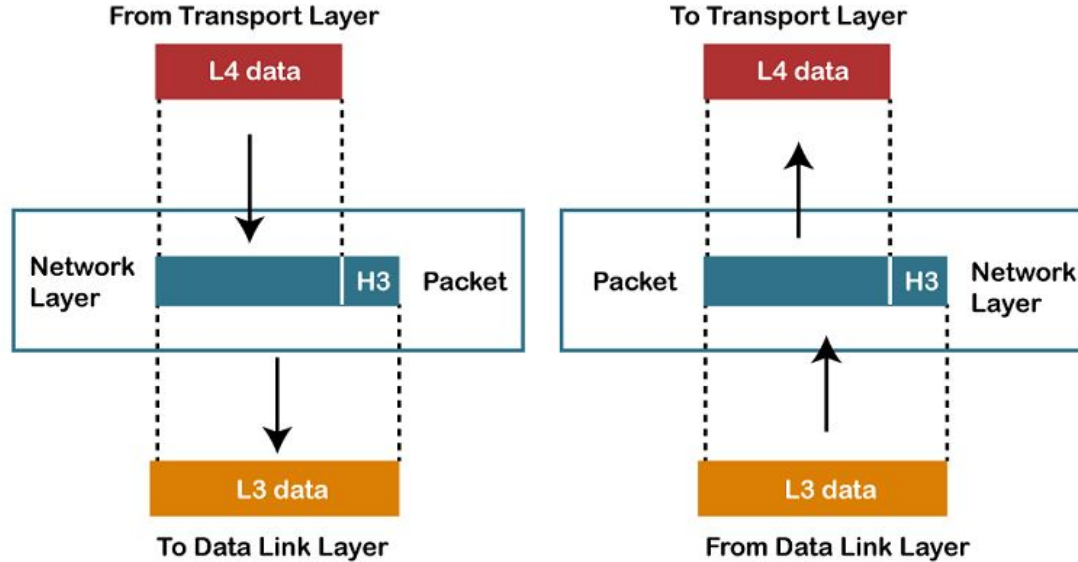
4.1

► Network Layer:

- ❖ The main aim of the Network Layer is the **source-to-destination delivery** of a packet across multiple networks (links). Whereas the data link layer oversees the delivery of a packet between two systems on the same network (links), the network layer ensures that **each packet gets from its source to its final destination**.
 - It also breaks the messages that have to be sent into **packets** and to assemble incoming packets into messages for higher levels.
 - If systems are attached to the same network, there is usually no need for a network layer.
 - However, if two systems are attached connecting devices on the **different networks** (links), so there is often needed for the network layer to complete the source-to-destination delivery of the message.

► Network Layer:

- ❖ This figure shows the relationship of the network layer to the data link and transport layer:



► Functions of Network Layer:

❖ Routing:

- When a packet reaches the **router's input link**, the router will move the packets to the **router's output link**. For example, a packet from S1 to R1 must be forwarded to the next router on the path to S2.

❖ Logical Addressing:

- The data link layer implements the physical addressing and network layer implements the **logical addressing**. Logical addressing is also used to distinguish between source and destination system. The network layer adds a header to the packet which includes the logical addresses of both the sender and the receiver.

❖ Internetworking:

- This is the **main role of the network layer** that it provides the logical connection between different types of networks.

► Functions of Network Layer:

❖ Fragmentation and Path determining ::

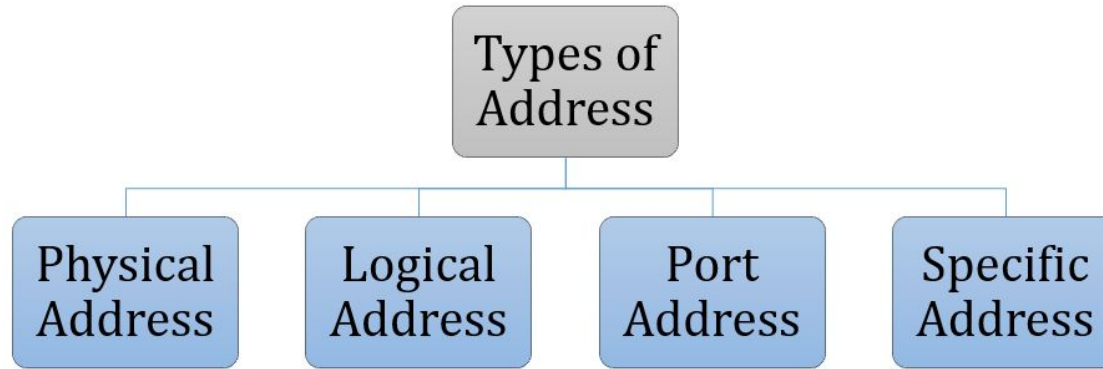
- The fragmentation is a process of **breaking the packets** into the smallest individual data units that travel through different networks. And determine best path among multiple paths.



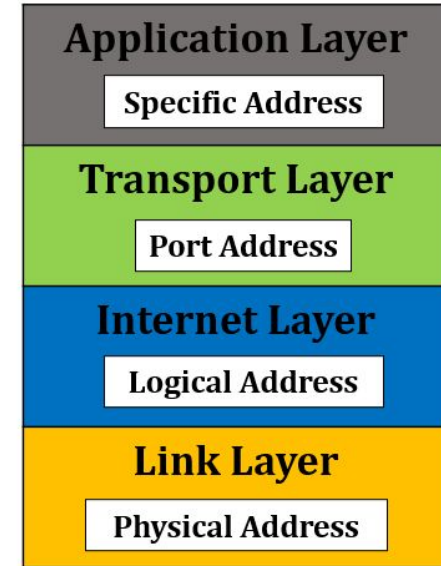
▶ **IPV4 Addressing & Subnetting**

4.2

► Types of Address:



TCP/IP Model



► Physical Addressing:

- ❖ The physical address, also known as the **link address**, is the address of a node as defined by its LAN or WAN.
- ❖ It is **included in the frame** used by the data link layer. It is the lowest-level address.
- ❖ The physical addresses have authority **over the network** (LAN or WAN).
- ❖ The size and format of these addresses vary depending on the network.
 - For example, Ethernet uses a 6-byte (48-bit) physical address that is imprinted on the network interface card (NIC).
 - 07:01:02:01:2C:4B
 - A 6-byte (12 hexadecimal digits) physical address

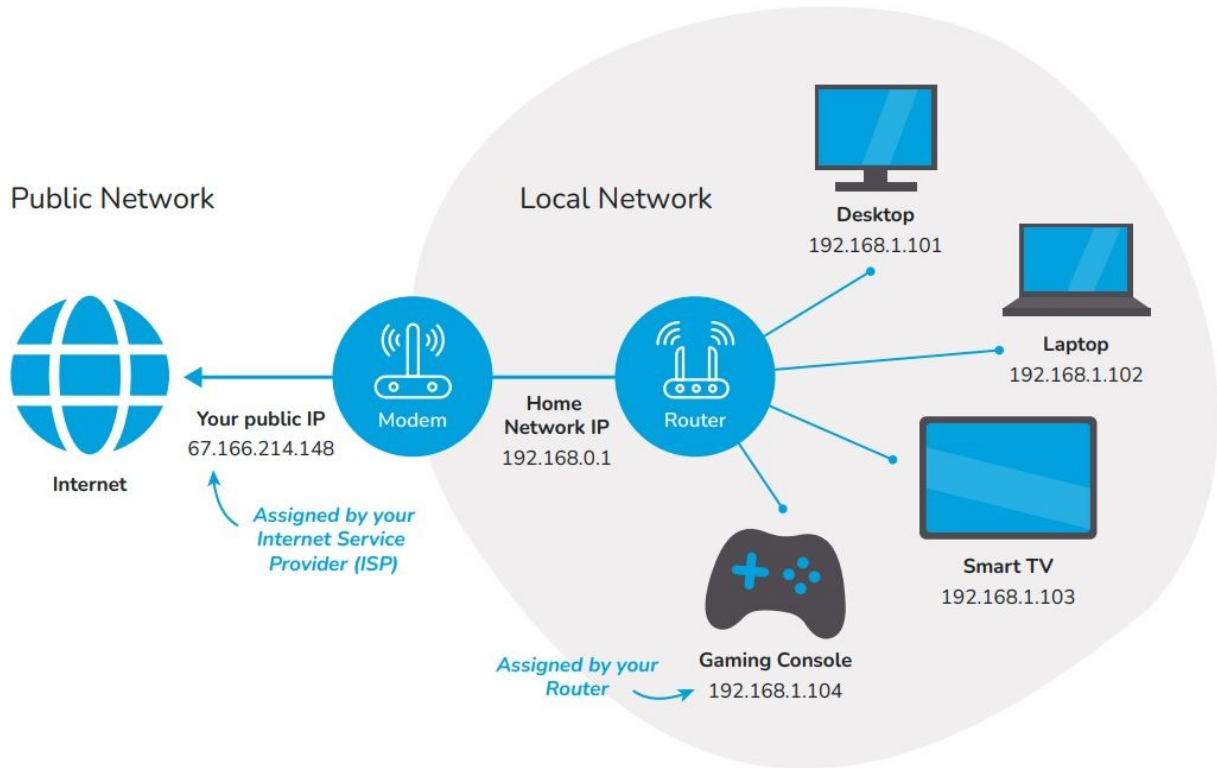
► Logical Addressing:

- ❖ Logical addresses are necessary for **universal communications** that are independent of underlying physical networks.
- ❖ A logical address in the Internet is **currently a 32-bit address** that can uniquely define a host connected to the Internet.
- ❖ **No two** publicly addressed and visible hosts on the Internet can have the **same** IP address.
- ❖ A logical address contains **two components**:
 - **Network ID** – identifies which network a host belongs to.
 - **Host ID** – uniquely identifies the host on that network.

► Port Addressing:

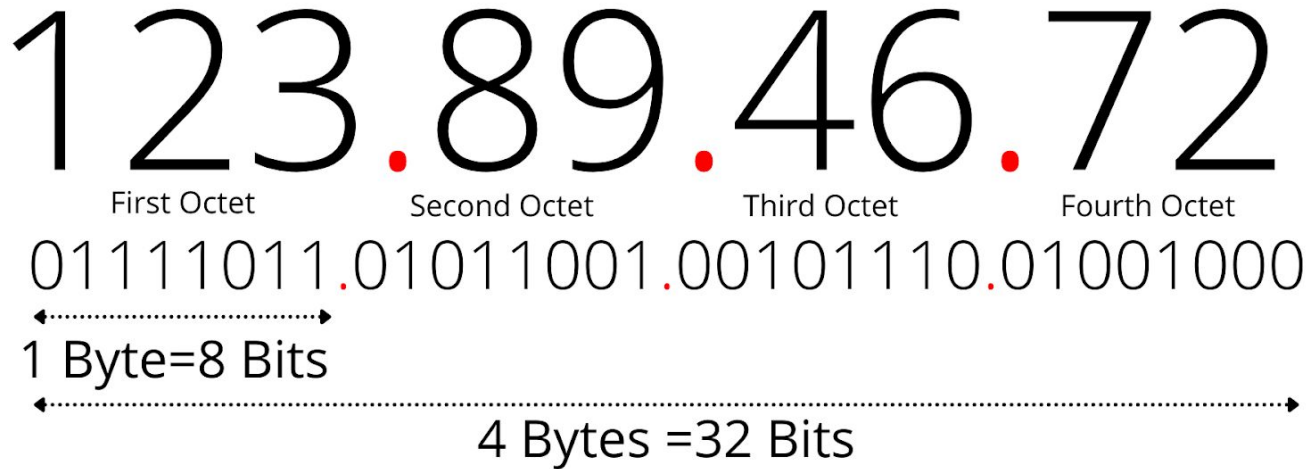
- ❖ The IP address and the physical address are necessary for a quantity of data to travel from a source.
- ❖ Today, computers are devices that can run multiple processes at the same time.
- ❖ For these processes to receive data simultaneously, we need a method to **label the different processes**. In other words, they need addresses.
- ❖ In the TCP/IP architecture, the **label assigned to a process** is called a port address.
- ❖ A port address in TCP/IP is **16 bits** in length.

► IP Addressing:



► IP Addressing:

IPv4 Address Format (Dotted Decimal Notation)



► IP Addressing (Logical Addressing):

- ❖ **What is an IP Address?** An IP address is a **numerical value/label** assigned to each device connected to a **computer network** that uses the Internet Protocol for communication.
- ❖ There are **two versions** of IP addresses:
 - IPv4 (32-bit) and IPv6 (128-bit). However, IPv4 is still more widely used.
- ❖ IP addressing is a function of the Network Layer of the OSI Model.
- ❖ An IP address contains **two components**:
 - **Network ID** – identifies which network a host belongs to(is connected).
 - **Host ID** – uniquely identifies the host on that network.
- ❖ Examples of logical addressing protocols include **Internetwork Packet Exchange (IPX)** and **Internet Protocol (IP)**. IPX was predominantly used on Novell networks, but is now almost entirely deprecated. IP is the most widely-used logical address, and is the backbone protocol of the Internet.

► Types of IP address:

Four types of IP Addresses

Public



Private



Static



Dynamic



► Types of IP Address:

- **Public IP Address:** A public IP address is an address where **one primary address** is associated with **your whole network**. In this type of IP address, **each** of the connected devices has the **same IP address**. This type of public IP address is provided to your router by your ISP.
- **Private IP Address:** A private IP address is a **unique IP number** assigned to every device that connects to your home internet network, which includes devices like computers, tablets, smartphones, which is used in your household.

► Types of IP Address:

- **Dynamic IP Address:** Dynamic IP addresses **always keep changing**. It is temporary and are allocated to a device every time it connects to the web. Dynamic IPs can trace their origin to a collection of IP addresses that are shared across many computers. A dynamic IP address will be assigned by a **Dynamic Host Configuration Protocol (DHCP) server**, which is subject to change.
- **Static IP Addresses :** A static IP address is an IP address that **cannot be changed**. Static IP address never changes, but it can be altered as part of routine network administration.

► Versions of IP Address:

Versions of IP Addresses



IPv4

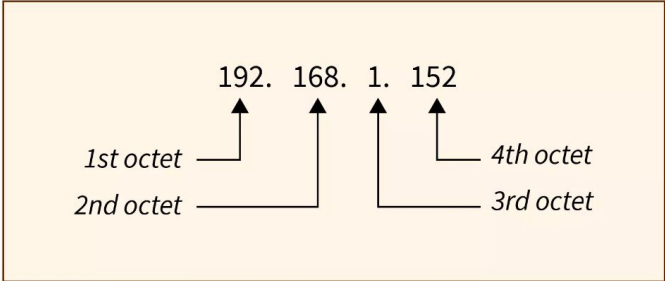
IPv6

► IPv4 Addressing:

- IPv4 was the **first version of IP**. It was deployed for production in the ARPANET in 1983.
- Today it is the most widely used IP version. It is used to **identify devices on a network** using an addressing system.
- An IPv4 address is **32 bits** long.
- The IPv4 addresses are unique and universal .
- IPv4 uses 32-bit addresses, which means that the **address space is 2^{32}** or 4,294,967,296 (Maximum available theoretically) .
- A core function of IP is to provide logical addressing for hosts.
- An IP address provides a **hierarchical structure** to both uniquely identify a host, and what network that host exists on.

► IPv4 Addressing:

- The IPv4 address is divided into **two parts** - the network part and the host part(also referred to as **netid** and **hostid**).



192. 168. 1. 152

1st octet 2nd octet 3rd octet 4th octet

The diagram shows the IP address 192.168.1.152 with arrows pointing from labels to each octet: '1st octet' to 192, '2nd octet' to 168, '3rd octet' to 1, and '4th octet' to 152.

Example IP: 192.168.1.152

Dotted Decimal	75.	45.	34.	78.
Binary	01001011	00101101	00100010	01001110
Hexadecimal	4B	2D	22	4E

Different Notations of IP Address

► IPv4 Addressing:

IPv4 Addressing

A core function of IP is to provide logical addressing for hosts. An **IP address** provides a hierarchical structure to both uniquely identify a *host*, and what *network* that host exists on.

An IP address is most often represented in **decimal**, in the following format:

158.80.164.3

An IP address is comprised of four **octets**, separated by periods:

First Octet	Second Octet	Third Octet	Fourth Octet
158	80	164	3

Each octet is an **8-bit** number, resulting in a **32-bit IP address**. The smallest possible value of an octet is 0, or 00000000 in binary. The largest possible value of an octet is 255, or 11111111 in binary.

The above IP address represented in binary would look as follows:

First Octet	Second Octet	Third Octet	Fourth Octet
10011110	01010000	10100100	00000011

► Characteristics of IPV4 Addressing:

- IPv4 addresses are **32-bit long**.
- They are either represented in **binary, dotted-decimal, or hexadecimal notation**. The most common form to represent IPv4 addresses is the dotted decimal notation.
- IPv4 addresses are **classified into classful addressing** where the address space is divided into five classes- Class A, Class B, Class C, Class D, and Class E.
- IPv4 addresses are **unique**, so two devices on a network can never have the same IP address.
- IPv4 address consists of **two parts** - the network part and the host part.
- The **IPv4 packet header** consists of 20 bytes of data and the number of the header field is 12.
- IPv4 is a **connectionless protocol**.
- IPv4 has **3 modes of addressing**- unicast, broadcast, and multicast.
- IPv4 can be **assigned** manually or by a protocol known as DHCP(Dynamic Host Configuration Protocol).
- IPv4 can be unreliable while transmitting packets.

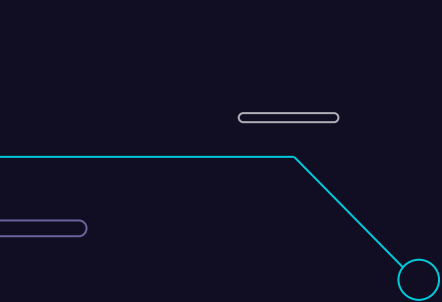
► Advantages of IPV4 Addressing:

- IPv4 is a connectionless protocol.
- The IPv4 routing can be handled easily by all the systems.
- Across a large network, **IPv4 can connect various devices** and along with connection, the verification can also be done. This is done with the use of NAT(Network Address Translation).
- The process of **routing is carried out smoothly** because the addresses are combined more effectively.
- **Privacy and security** are maintained in IPv4 as the data is encrypted in the packets.
- The encoding in IPv4 is flawless.

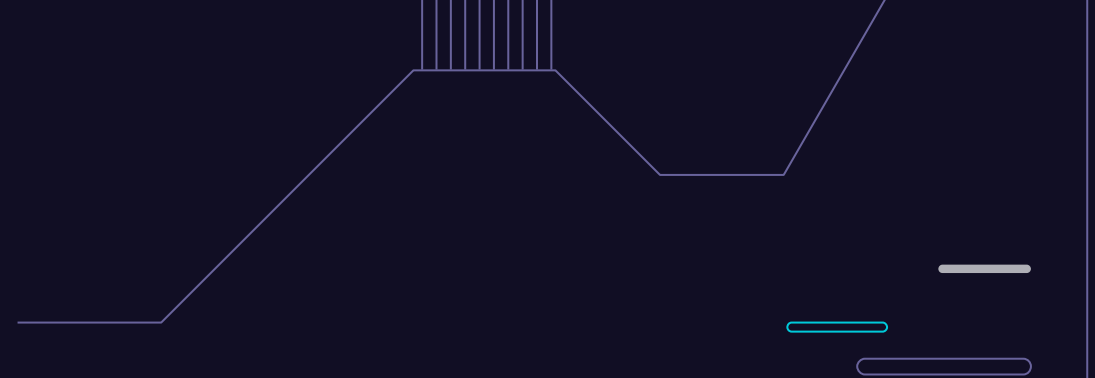
► Disadvantages of IPV4 Addressing:

- IPv4 can be assigned manually or by a protocol known as DHCP and if it is done through DHCP, it needs a **lot of management** for its infrastructure.
- Since IPv4 was established way back, its implementation did not provide security against **threats introduced today**. Internet Protocol Security (IPSec) enables network security to IPv4 by specifying the use of the internet. However the problem arises when **IPSec is not built-in** and its implementation is optional.
- Most of the IP addresses are **reserved in the United States**.

• **To overcome the drawbacks of IPv4, IPv6 was introduced.**



► **Classful & Classless Addressing**



4.3

► Need for Classful Addressing:

- Initially in 1980's IP address was divided into two fixed part i.e.,
 - NID(Network ID) = 8bit, and
 - HID(Host ID) = 24bit.
- So there are 2^8 that is **256 total network** are created and 2^{24} that is **16M Host per network**.
- There are one 256 Networks and even a small organization must buy 16M computer (Host) to purchase one network. That's why we need classful addressing.

► Classful Addressing:

- Classful Addressing is an addressing method **that divides** the IP address into **five separate classes** based on four address bits.
- A legacy IP addressing scheme that divides IP addresses into **five fixed classes** (A, B, C, D, and E).

Key Characteristics:

- Uses **fixed-length subnet masks** (e.g., 255.0.0.0 for Class A, 255.255.0.0 for Class B).
- Subnetting is not flexible, as it's limited by the **class boundaries**.
- Leads to **potential address waste** due to fixed allocation sizes.
- Does not support **Classless Inter-Domain Routing (CIDR)**.

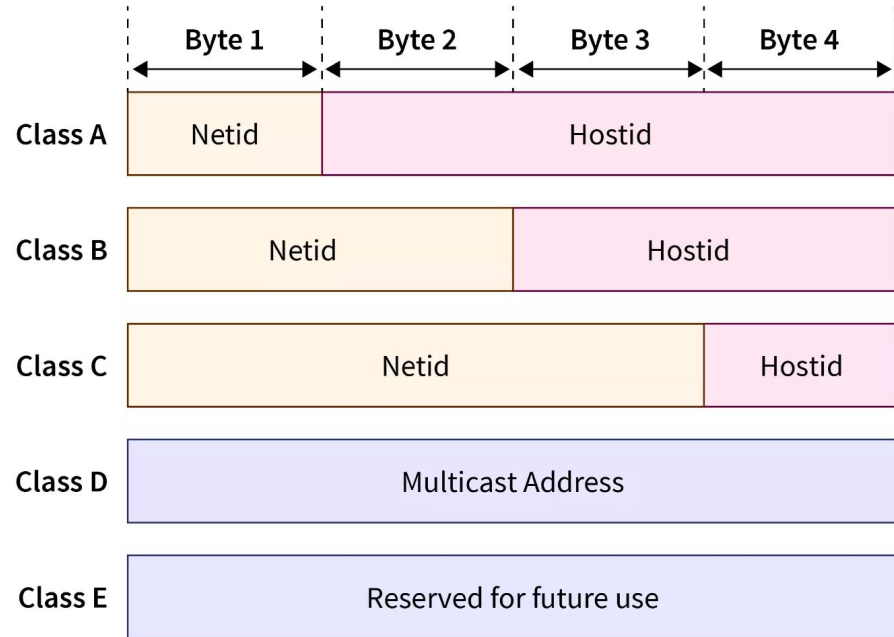
► Classful Addressing:

An IP address is 32-bit long. An IP address is divided into subclasses:

- Class A
- Class B
- Class C
- Class D
- Class E

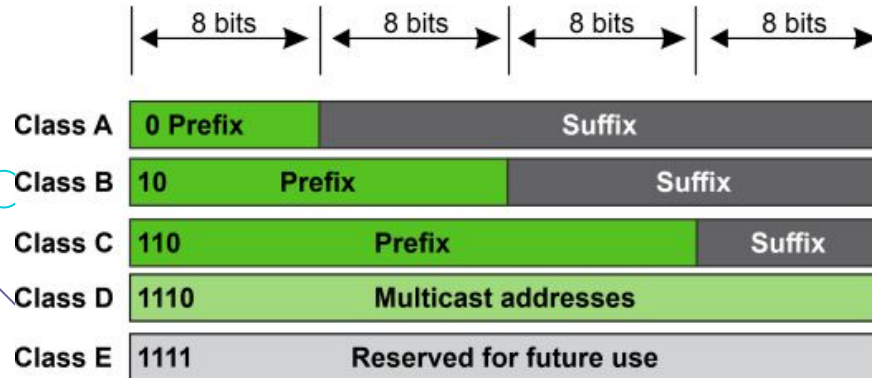
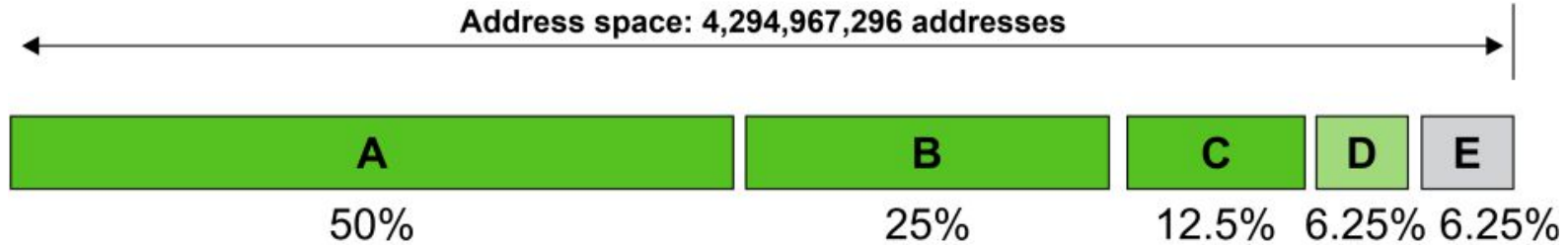
An ip address is divided into two parts:

- **Network ID:** It represents the number of networks.
- **Host ID:** It represents the number of hosts.



► Classful Addressing:

Occupation of the address space in classful addressing



Classes	Prefixes	First Byte
A	n = 8 bits	0 to 127
B	n = 16 bits	128 to 191
C	n = 24 bits	192 to 223
D	Not applicable	224 to 239
E	Not applicable	240 to 255

► IP Address Classes:

- ❖ The IPv4 address space has been structured into several classes. The **value of the first octet of an address** determines the class of the network:

Five Different Classes of IPv4 Addresses

Class	First Octet decimal (range)	First Octet binary (range)	IP range	Subnet Mask	Hosts per Network ID	# of networks
Class A	0 — 127	0 XXXXXXX	0.0.0.0-127.255.255.255	255.0.0.0	$2^{24}-2$	2^7
Class B	128 — 191	10 XXXXXX	128.0.0.0-191.255.255.255	255.255.0.0	$2^{16}-2$	2^{14}
Class C	192 — 223	110 XXXXX	192.0.0.0-223.255.255.255	255.255.255.0	2^8-2	2^{21}
Class D (Multicast)	224 — 239	1110 XXXX	224.0.0.0-239.255.255.255			
Class E (Experimental)	240 — 255	1111 XXXX	240.0.0.0-255.255.255.255			

► Class A Networks:

- ❖ This IP address class is used when there are a **large number of hosts**.
- ❖ In a Class A type of network, the **first 8 bits** (also called the first octet) identify the network, and the **remaining have 24 bits** for the host into that network.

- The network ID is 8 bits long.
- The host ID is 24 bits long.
- Class A IP address format is thus: **0NNNNNNN.HHHHHHHH.HHHHHHHH.HHHHHHHH**

00000001 - 01111111
1 - 127

- ❖ In Class A, the first bit in higher order bits of the first octet is **always set to 0** and the remaining 7 bits determine the network ID. The 24 bits determine the host ID in any network.
- ❖ The **total number of networks** in Class A = $2^7 = 128$ network address
- ❖ The **total number of hosts** in Class A = $2^{24} - 2 = 16,777,214$ host address
- ❖ **Example:** Class A address is 102.168.212.226. Here, "102" helps you identify the network and 168.212.226 identify the host.
- ❖ Subnet Mask: 255.0.0.0
 - 127.X.X.X is reserved for loopback address
 - 0.X.X.X is reserved for default network



► Class B Networks:

- ❖ In Class B, an IP address is assigned to those networks that **range from small-sized to large-sized** networks.

- The Network ID is **16 bits long**.

- The Host ID is **16 bits long**.

- Class B IP address format is: **10NNNNNN.NNNNNNN.NNNNNNN.NNNNNNN**

10000000 – 10111111
128 – 191

- ❖ In Class B, the higher order bits of the first octet is **always set to 10**, and the remaining 14 bits determine the network ID. The other 16 bits determine the Host ID.

- The **total number of networks** in Class B = $2^{14} = 16384$ network address

- The **total number of hosts** in Class B = $2^{16} - 2 = 65534$ host address

- ❖ Class B networks range from 128 to 191.

- ❖ **Example** of Class B IP address is 168.212.226.204, where *168.212* identifies the network and *226.204* helps you identify the network host.

- ❖ Subnet Mask: 255.255.0.0

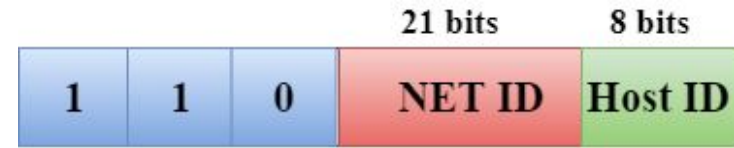
14 Bit		16 Bit	
1	0	Network	Host

Class B

► Class C Networks:

- ❖ Class C is a type of IP address that is used for the **small network**.
 - The **Network ID** is 24 bits long.
 - The **Host ID** is 8 bits long.
 - Class C IP address format is: **110NNNNN.NNNNNNNN.NNNNNNNN.HHHHHHHH**
- ❖ In Class C, the higher order bits of the first octet is **always set to 110**, and the remaining 21 bits determine the network ID. The 8 bits of the host ID determine the host in a network.
 - The **total number of networks** = $2^{21} = 2097152$ network address
 - The **total number of hosts** = $2^8 - 2 = 254$ host address
- ❖ Class C networks range from 192 to 223.
- ❖ **Example of a Class C address:**
 - Address: 207.79.233.6
 - Subnet Mask: 255.255.255.0

11000000 – **110**11111
192 – 223



► Class D Networks:

- ❖ Class D addresses are only used for **multicasting applications**. Class D is **never** used for regular networking operations. This class addresses the **first three bits set to "1"** and their **fourth bit set to use for "0"**. Class D addresses are 32-bit network addresses. All the values within the range are **used to identify multicast groups uniquely**.
- ❖ Therefore, there is no requirement to extract the host address from the IP address, so Class D **does not have** any subnet mask.
- ❖ **Example for a Class D IP address:**
 - 227.21.6.173

11100000 – **1110**1111
224 – 239



► Class E Networks:

- ❖ Class E IP address is defined by including the starting **four network address bits as 1**, which allows you to incorporate addresses from 240.0.0.0 to 255.255.255.255. However, **E class is reserved, and its usage is never defined**. Therefore, many network implementations discard these addresses as undefined or illegal.

- ❖ **Example for a Class E IP address:**

- 243.164.89.28



► Find the class of each address:

1. 00000001 00001011 00001011 11101111
2. 11000001 10000011 00011011 11111111
3. 14.23.120.8
4. 252.5.15.111

► Answers:

1. The first bit is 0. This is a class A address.
2. The first 2 bits are 1 1; the third bit is 0 (i.e 110). This is a class C address.
3. The first byte is 14 (between 0 and 127); the class is A.
4. The first byte is 252 (between 240 and 255); the class is E

► Limitations of Classful Addressing:

- Risk of **running out of address** space soon
- **Class boundaries** did not encourage efficient allocation of address space

Class	Higher bits	NET ID bits	HOST ID bits	No.of networks	No.of hosts per network	Range
A	0	8	24	2^7	2^{24}	0.0.0.0 to 127.255.255.255
B	10	16	16	2^{14}	2^{16}	128.0.0.0 to 191.255.255.255
C	110	24	8	2^{21}	2^8	192.0.0.0 to 223.255.255.255
D	1110	Not Defined	Not Defined	Not Defined	Not Defined	224.0.0.0 to 239.255.255.255
E	1111	Not Defined	Not Defined	Not Defined	Not Defined	240.0.0.0 to 255.255.255.255

► Classless Addressing:

- ❖ There is **no classes hierarchy** in the IP address but address is still granted in blocks.
- ❖ A **more flexible and efficient** IP addressing scheme that doesn't rely on fixed classes. It uses **variable-length subnet masks (VLSMs)** to divide IP address blocks into subnets of varying sizes.
- ❖ In modern networking, classless addressing and VLSM are **generally preferred** due to their efficiency and flexibility in IP address allocation and routing.
- ❖ **Formal Name: CIDR (Classless Inter Domain Routing):**
CIDR or Class Inter-Domain Routing was introduced in 1993 **to replace** classful addressing. Classless Inter-Domain Routing (CIDR) is a method for **efficiently allocating IP addresses and routing** Internet Protocol (IP) packets.

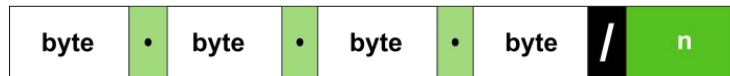
► Classless Inter Domain Routing Notation:

- ❖ CIDR IP Addresses look like-
 - a.b.c.d / n
- ❖ They **end with a slash** followed by a number called as **IP network prefix**.
 - IP network prefix tells the **number of bits** used for the identification of network.
 - **Remaining bits** are used for the identification of hosts in the network.

Example:

- ❖ An example of CIDR IP Address is-
 - 182.0.1.2 / 28
- ❖ It suggests- **28 bits** are used for the identification of network.
- ❖ Remaining **4 bits** are used for the identification of hosts in the network.

Slash notation (CIDR)



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Prefix
length

Examples:

12.24.76.8/8
23.14.67.92/12
220.8.24.255/25

► Classful vs Classless Addressing:

Parameter	Classful Addressing	Classless Addressing
Basics	In Classful addressing IP addresses are allocated according to the classes- A to E.	Classless addressing came to replace the classful addressing and to handle the issue of rapid exhaustion of IP addresses.
Practical	It is less practical.	It is more practical.
Network ID and Host ID	The changes in the <u>Network ID and Host ID</u> depend on the class.	There is no such restriction of class in classless addressing.
VLSM	It does not support the Variable Length Subnet Mask (VLSM).	It supports the Variable Length Subnet Mask (VLSM).
Bandwidth	Classful addressing requires more bandwidth. As a result, it becomes slower and more expensive as compared to classless addressing.	It requires less bandwidth. Thus, fast and less expensive as compared to classful addressing.
CIDR	It does not support <u>Classless Inter-Domain Routing (CIDR)</u> .	It supports Classless Inter-Domain Routing (CIDR).
Updates	Regular or periodic updates	Triggered Updates
Troubleshooting and Problem detection	Troubleshooting and problem detection are easy than classless addressing because of the division of network, host and subnet parts in the address.	It is not as easy compared to classful addressing.
Division of Address	• Network • Host • Subnet	• Host • Subnet

The background features a dark blue field with white and light blue geometric lines resembling a circuit board. These lines include straight paths, right-angle turns, and small circles at various points. A prominent light blue line starts from the left edge, goes down, and then right, ending in a small circle. Other lines are scattered across the top and right sides, some ending in small horizontal bars or circles.

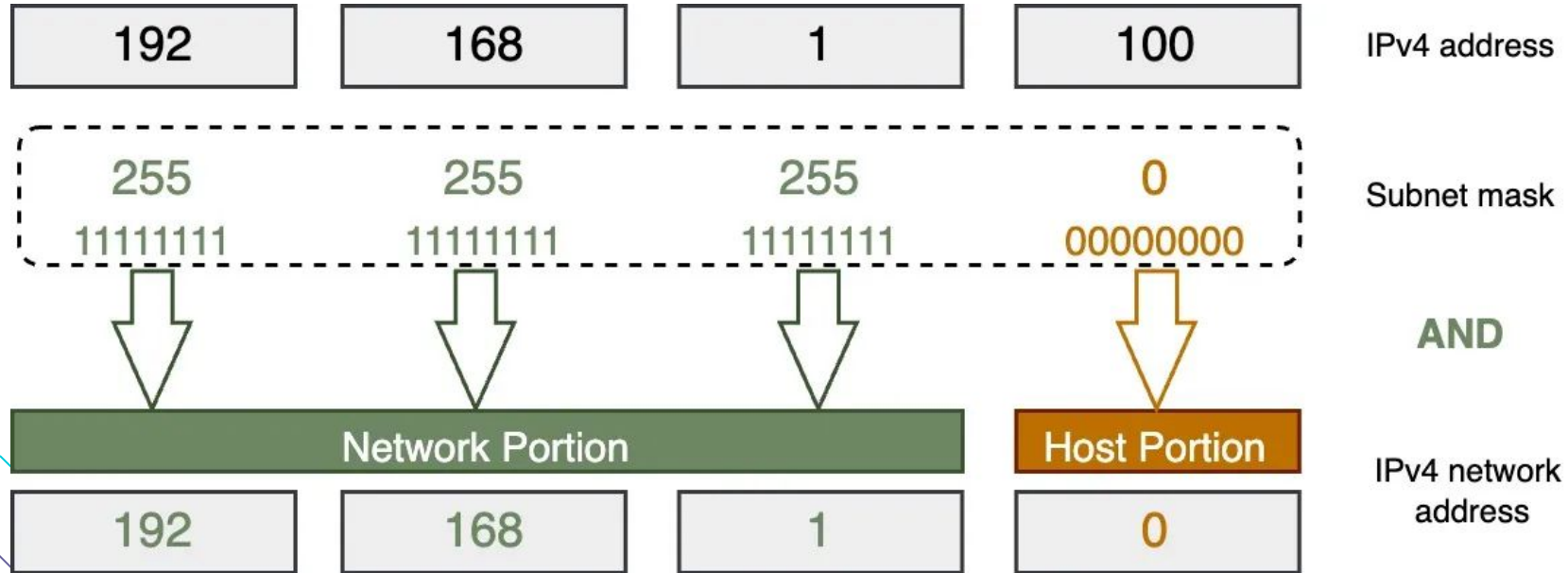
► Subnetting & Supernetting

4.4

► Subnet Mask:

- ❖ A subnet mask is a **32 bits address** used to distinguish between a network address and a host address in IP address.
- ❖ The subnet mask determines **what part of an address** identifies the network, and what part identifies the host.
- ❖ Each class has a **default subnet mask**. A network using its default subnet mask is referred to as a class-full network.
- ❖ For example, 10.1.1.1 is a Class A address, and its default subnet mask is 255.0.0.0 (/8 in CIDR).
- ❖ The subnet address is used to identify the network itself. A routing table contains a list of known networks, and each network is identified by its subnet address. Subnet addresses contain all 0 bits in the host portion of the address.

► Subnet Mask Values:



► Valid Subnet Mask Table:

Valid Subnet Mask Values

128	64	32	16	8	4	2	1	Decimal Value
0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	128
1	1	0	0	0	0	0	0	192
1	1	1	0	0	0	0	0	224
1	1	1	1	0	0	0	0	240
1	1	1	1	1	0	0	0	248
1	1	1	1	1	1	0	0	252
1	1	1	1	1	1	1	0	254
1	1	1	1	1	1	1	1	255

► Valid Subnet Mask Table- **CIDR:**

<i>/n</i>	<i>Mask</i>	<i>/n</i>	<i>Mask</i>	<i>/n</i>	<i>Mask</i>	<i>/n</i>	<i>Mask</i>
/1	128.0.0.0	/9	255.128.0.0	/17	255.255.128.0	/25	255.255.255.128
/2	192.0.0.0	/10	255.192.0.0	/18	255.255.192.0	/26	255.255.255.192
/3	224.0.0.0	/11	255.224.0.0	/19	255.255.224.0	/27	255.255.255.224
/4	240.0.0.0	/12	255.240.0.0	/20	255.255.240.0	/28	255.255.255.240
/5	248.0.0.0	/13	255.248.0.0	/21	255.255.248.0	/29	255.255.255.248
/6	252.0.0.0	/14	255.252.0.0	/22	255.255.252.0	/30	255.255.255.252
/7	254.0.0.0	/15	255.254.0.0	/23	255.255.254.0	/31	255.255.255.254
/8	255.0.0.0	/16	255.255.0.0	/24	255.255.255.0	/32	255.255.255.255

► Classful Uses:

- ❖ Class A $\Rightarrow$ /8 to /15
- ❖ Class A, B $\Rightarrow$ /16 to /23
- ❖ Class A, B, C $\Rightarrow$ /24 to /30

► Valid Subnet Mask Table:

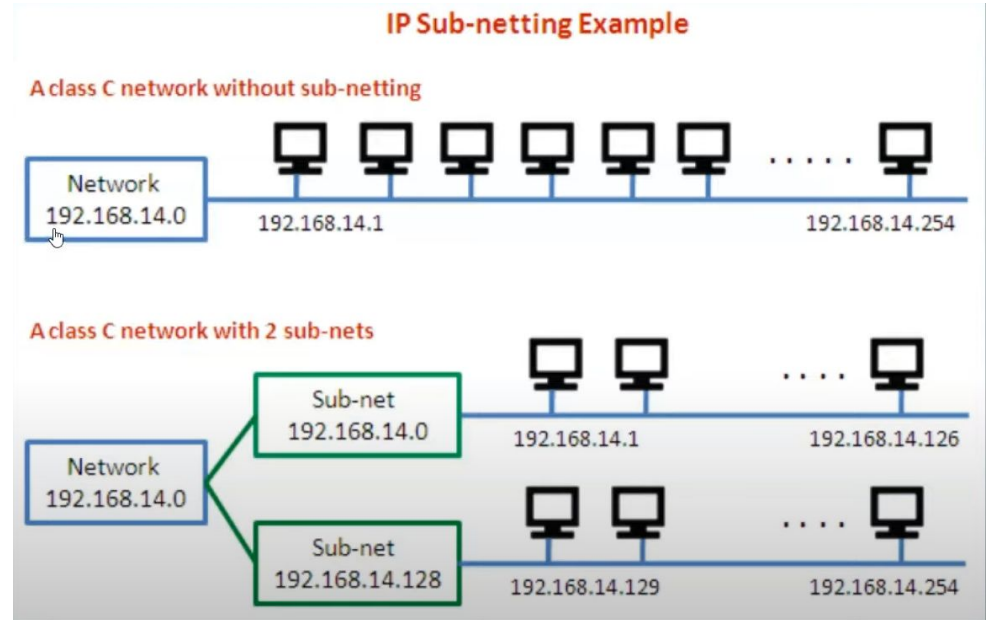
Subnet Mask (Decimal)	Subnet Mask (Binary)	Valid/ Invalid
255.255.255.240	1111111.1111111.1111111.11110000	Valid
255.230.255.0	1111111.11100110.1111111.00000000	Invalid
255.255.0.0	1111111.1111111.00000000.00000000	Valid
240.0.0.0	11110000.00000000.00000000.00000000	Valid
223.0.0.0	1101111.00000000.00000000.00000000	Invalid
255.0.255.0	1111111.00000000.1111111.00000000	Invalid

► Identify the invalid subnet masks:

- 255.240.0.0
- 248.0.0.0
- 255.255.125.0
- 255.255.255.252
- 255.255.242.0

► Subnetting:

- ❖ A subnetwork or subnet is a **logical division** of an IP network.
- ❖ Subnetting is a method of dividing a **single physical network** into logical sub-networks (subnets).
- ❖ The practice of dividing a network into two or more networks is called subnetting.



► Subnetting:

- ❖ Subnetting is the process of creating new networks (or subnets) by **stealing bits** from the **host portion** of a subnet mask. As a result, stealing bits from hosts **creates more networks** but **fewer hosts** per network.
- ❖ Consider the following Class C network:
 - 192.168.254.0
- ❖ The default subnet mask for this network is 255.255.255.0. This single network can be segmented, or sub-netted, into multiple networks.
- ❖ For example, assume a minimum of 10 new networks are required. Resolving this is possible using the following magical formula: 2^n .

► Subnetting:

- ❖ The exponent 'n' **identifies** the number of bits to steal from the host portion of the subnet mask. The default Class C mask (255.255.255.0) looks as follows in binary:
 - 11111111.11111111.11111111.00000000
- ❖ There are a total of 24 bits set to 1, which are used to identify the network. There are a total of 8 bits set to 0, which are used to identify the host, and these host bits can be stolen.
- ❖ Stealing bits essentially involves **changing host bits (set to 0 or off)** in the subnet mask to network bits (set to 1 or on). Remember, network bits in a subnet mask must always be contiguous - **skipping bits is not allowed.**

► Subnetting:

- ❖ Consider the result if **three bits are stolen**. Using the above formula:
 - $2^n = 2^3 = 8 = 8$ new sub-networks created.
- ❖ However, a total of 8 new networks does not meet the original requirement of at least 10 networks. Consider the result if **four bits are stolen**:
 - $2^n = 2^4 = 16 = 16$ new networks created.
- ❖ A total of 16 new networks does meet the original requirement. Stealing four host bits results in the following new subnet mask:
 - 11111111.11111111.11111111.11110000 = 255.255.255.240

► The $2^n - 2$ Rule and Subnetted Networks:

- ❖ To avoid confusion, it was historically unacceptable to use the **first and last** new networks created when sub-netting, as it is possible for a class full network to have the same subnet and broadcast address as its sub-netted networks.
- ❖ This required the **2^n** formula to also be used when calculating the number of new networks created while sub-netting.
- ❖ Note: the formula for calculating usable hosts is always **$2^n - 2$** .

► What is the Use of Subnetting?

1. **Efficient IP Address Allocation:** Subnetting allows a single network block to be divided into smaller subnets, efficiently allocating IP addresses for various physical networks and managing limited address space.
2. **Improved Network Speed:** By breaking a large network into smaller subnets, traffic is contained within each subnet, reducing unnecessary network traffic and enhancing speed.
3. **Enhanced Network Security:** Subnetting isolates network segments, which helps contain and quarantine security threats and makes it harder for unauthorized users to access the entire network.
4. **Reduced Network Congestion:** Grouping related devices into the same subnet minimizes the visibility of data packets across the network, reducing overall congestion.
5. **Increased Efficiency:** Subnetting helps streamline network traffic by reducing the need for additional routers and avoiding unnecessary data detours, speeding up data transmission.

► Subnetting- 5 Steps:

1. Identify the class of IP address and note the Default Subnet Mask.
2. Convert the default Subnet Mask into Binary.
3. Note the number of hosts required per subnet and find the Subnet Generator(SG) and octet position
4. Generate the new subnet mask.
5. Use the SG and generate the network ranges (Subnets) in the appropriate octet position.

► Subnetting in **Class C** Addressing:

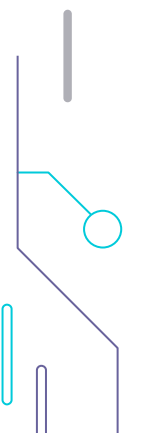
Network Bits	Subnet Mask	Bits Borrowed	Subnets	Hosts/Subnet
24	255.255.255.0	0	1	254
25	255.255.255.128	1	2	126
26	255.255.255.192	2	4	62
27	255.255.255.224	3	8	30
28	255.255.255.240	4	16	14
29	255.255.255.248	5	32	6
30	255.255.255.252	6	64	2

Note:

- ❖ Class A => /8 to /15
- ❖ Class A, B => /16 to /23
- ❖ Class A, B, C => /24 to /30



▶ **Q: If 200.100.10.66/26 is IPV4 address then answer the following questions.**

1. Is this host, network or broadcast address?
 2. What is the subnet mask in dotted decimal?
 3. What is the network address?
 4. What is the broadcast address?
 5. What is the first usable host address ?
 6. What is the last usable host address?
 7. How many usable host are there in the network?
- 



▶ **Q: If 200.100.10.66/26 is IPV4 address then answer the following questions.**



Homework:



Q: If 200.17.21.128/27 is IPV4 address then answer the following questions.

1. Is this host, network or broadcast address?
2. What is the subnet mask in dotted decimal?
3. What is the network address?
4. What is the broadcast address?
5. What is the first usable host address ?
6. What is the last usable host address?
7. How many usable host are there in the network?

► Subnetting in Class B Addressing:

Note:

- ❖ Class A => /8 to /15
- ❖ Class A, B => /16 to /23
- ❖ Class A, B, C => /24 to /30

Network Bits	Subnet Mask	Bits Borrowed	Subnets	Hosts/Subnet
16	255.255.0.0	0	0	65534
17	255.255.128.0	1	2	32766
18	255.255.192.0	2	4	16382
19	255.255.224.0	3	8	8190
20	255.255.240.0	4	16	4094
21	255.255.248.0	5	32	2046
22	255.255.252.0	6	64	1022
23	255.255.254.0	7	128	510
24	255.255.255.0	8	256	254
25	255.255.255.128	9	512	126
26	255.255.255.192	10	1024	62
27	255.255.255.224	11	2048	30
28	255.255.255.240	12	4096	14
29	255.255.255.248	13	8192	6
30	255.255.255.252	14	16384	2

► **Q: If 172.16.0.0/17 is IPV4 address then answer the following questions.**

1. Is this host, network or broadcast address?
2. What is the subnet mask in dotted decimal?
3. What is the network address?
4. What is the broadcast address?
5. What is the first usable host address ?
6. What is the last usable host address?
7. How many usable host are there in the network?



▶ **Q: If 172.16.0.0/17 is IPV4 address then answer the following questions.**





Homework:

Q: If 180.34.64.64/26 is IPV4 address then answer the following questions.

1. Is this host, network or broadcast address?
2. What is the subnet mask in dotted decimal?
3. What is the network address?
4. What is the broadcast address?
5. What is the first usable host address ?
6. What is the last usable host address?
7. How many usable host are there in the network?

► Subnetting in **Class A** Addressing:

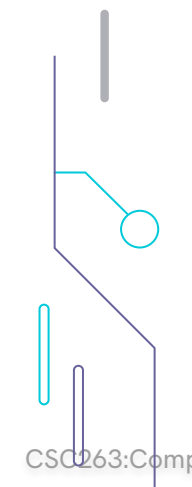
Note:

- ❖ Class A => /8 to /15
- ❖ Class A, B => /16 to /23
- ❖ Class A, B, C => /24 to /30

Network Bits	Subnet Mask	Bits Borrowed	Subnets	Hosts/Subnet
8	255.0.0.0	0	1	16777214
9	255.128.0.0	1	2	8388606
10	255.192.0.0	2	4	4194302
11	255.224.0.0	3	8	2097150
12	255.240.0.0	4	16	1048574
13	255.248.0.0	5	32	524286
14	255.252.0.0	6	64	262142
15	255.254.0.0	7	128	131070
16	255.255.0.0	8	256	65534
17	255.255.128.0	9	512	32766
18	255.255.192.0	10	1024	16382
19	255.255.224.0	11	2048	8190
20	255.255.240.0	12	4096	4094
21	255.255.248.0	13	8192	2046
22	255.255.252.0	14	16384	1022
23	255.255.254.0	15	32768	510
24	255.255.255.0	16	65536	254
25	255.255.255.128	17	131072	126
26	255.255.255.192	18	262144	62
27	255.255.255.224	19	524288	30
28	255.255.255.240	20	1048576	14
29	255.255.255.248	21	2097152	6
30	255.255.255.252	22	4194304	2



▶ **Q: If 10.1.0.0/9 is IPV4 address then answer the following questions.**

1. Is this host, network or broadcast address?
 2. What is the subnet mask in dotted decimal?
 3. What is the network address?
 4. What is the broadcast address?
 5. What is the first usable host address ?
 6. What is the last usable host address?
 7. How many usable host are there in the network?
- 



▶ **Q: If 10.1.0.0/9 is IPV4 address then answer the following questions.**





Homework:

Q: If 17.34.16.0/23 is IPV4 address then answer the following questions.

1. Is this host, network or broadcast address?
2. What is the subnet mask in dotted decimal?
3. What is the network address?
4. What is the broadcast address?
5. What is the first usable host address ?
6. What is the last usable host address?
7. How many usable host are there in the network?



► **Q: Subnet the IP address 216.21.5.0 into 30 hosts in each subnet.**





► **Q: A company is provided with the site address of 201.70.64.0.
The company needs 6 subnets. Design the subnets.**



► **Q. Is 192.16.144.64/27 a host, network, or broadcast address?
which layer of OSI model do Hub, Switch and Router operate
on?**

In

► **Q. Write the subnet ID and broadcast address of each subnet if you divide a class C network (192.168.3.0 – 192.168.3.255) into 4 different subnets. What is the new subnet mask?**

► **Q. What subnet ID and broadcast address of each subnet if you divide a class B network (150.10.0.0 – 150.10.255.255) in 4 different subnets. What is the new subnet mask?**

► **Q. In a block of address, we know the IP Address of one host is 25.34.12.56/16. What are the first address(network address) and the last address (limited broadcast address) in this block?**



► **Q: Subnet the IP address 196.10.20.0 into 52 hosts in each subnet.**



► **Q: A block of addresses is granted to a small organization. We know that one of the addresses is 205.16.37.39/28. What is the first address in the block ?**

► For More Details:

- <https://www.scaler.com/topics/computer-network/subnetting/>
- <https://www.guru99.com/subnetting-subnet-mask.html>



▶ **IPV6 Addressing & its Features**

4.5

► IPv6 Addressing:

- IPv6 is designed to be the **successor** to IPv4.
- With an increasing Internet population, a limited IPv4 address space, issues with **NAT** and an **Internet of things**, the time has come to begin the transition to IPv6.
- IPv4 has a theoretical maximum of 4.3 billion addresses, plus private addresses in combination with NAT.
- IPv6 larger **128-bit** address space provides for 340 undecillion addresses ($340 * 10^{36}$).
- IPv6 fixes the limitations of IPv4 and includes additional enhancements, such as ICMPv6.
- IPv6 is represented using hexadecimal number system.

► IPv6 Addressing:

Group 1 Group 2 Group 3 Group 4 Group 5 Group 6 Group 7 Group 8

2001:0df8:00f2:0000:0000:06ee:0000:0f11

Group 1 Group 2 Group 3 Group 6 Group 7 Group 8

2001:0df8:00f2::06ee:0000:0f11

128 - Bits

2001:0db8:0000:0000:34f4:0000:0000:f3dd

48-Bits

16-Bits

64-Bits

Network Part

Subnet ID

Client ID

► IPv6 Address Types:

There are three types of IPv6 addresses:

- **Unicast Address:** This type of address identifies a **single network interface**. It is used for **one-to-one** communication, where data is sent from one source to one destination.
- **Multicast Address:** Multicast addresses identify a **group of interfaces**, and data sent to a multicast address is delivered to all members of the group. It is used for **one-to-many** communication .
- **Anycast Address:** Anycast addresses are assigned to **multiple interfaces**, and the data is delivered to the **nearest** (in terms of routing distance) of these interfaces. Anycast is typically used for distributed services where the **user is served by the nearest node** with the anycast address.
- **Note:** IPv6 **does not have** broadcast addresses.

► IPV6 Address Features:

- ❖ The successor of IPv4 is not designed to be **backward compatible**. Trying to keep the basic functionalities of IP addressing, IPv6 is redesigned entirely. It offers the following features:
- ❖ **IPSec**
 - Initially it was decided that IPv6 **must have IPSec** security, making it more secure than IPv4. This feature has now been made optional.
- ❖ **No Broadcast**
 - Though Ethernet/Token Ring are considered as broadcast network because they support Broadcasting, IPv6 **does not have** any broadcast support any more. It **uses multicast** to communicate with multiple hosts.
- ❖ **Anycast Support**
 - This is another characteristic of IPv6. IPv6 has introduced **Anycast mode of packet routing**. In this mode, multiple interfaces over the Internet are assigned same Anycast IP address. Routers, while routing, send the packet to the nearest destination.

► IPv6 Address Features:

❖ Mobility

- IPv6 was designed keeping mobility in mind. This feature enables hosts (such as mobile phone) to roam around in different geographical area and remain connected with the **same IP address**. The mobility feature of IPv6 takes advantage of auto IP configuration and Extension headers.

❖ Auto-configuration, Large Address Space, Better Header format, Extensibility

► Hexadecimal Number System:

- ❖ Before introducing IPv6 Address format, we shall look into Hexadecimal Number System.
- ❖ Hexadecimal is a positional number system that uses radix (base) of 16.
- ❖ To represent the values in readable format:

Decimal	Binary	Hexadecimal
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8
9	1001	9
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F

► IPv6 Address Representation:

- ❖ IPv6 addresses consist of **8 groups of 16-bit hexadecimal values** separated by colons (:).

IPv6 addresses have the following format:

aaaa:aaaa:aaaa:aaaa:aaaa:aaaa:aaaa:aaaa

- ❖ Each aaaa is a **16-bit hexadecimal value**, and each a is a 4-bit hexadecimal value.

Following is a sample IPv6 address:

3FFE:0000:0000:0001:0200:F8FF:FE75:50DF

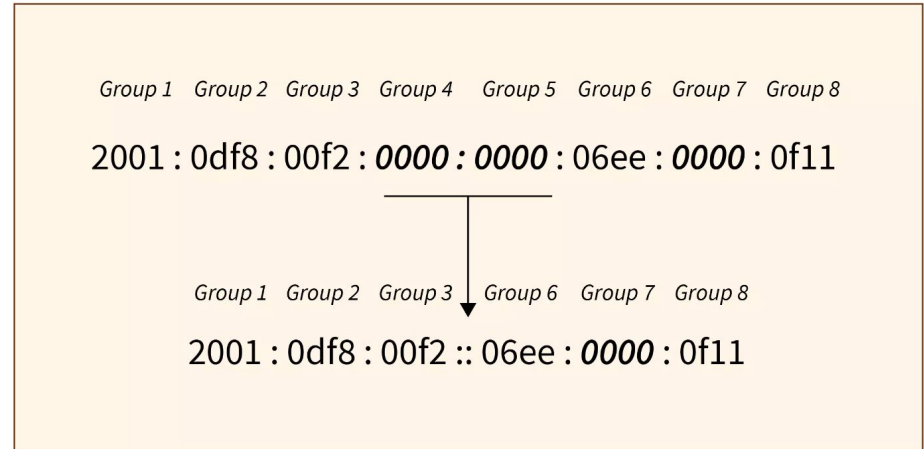
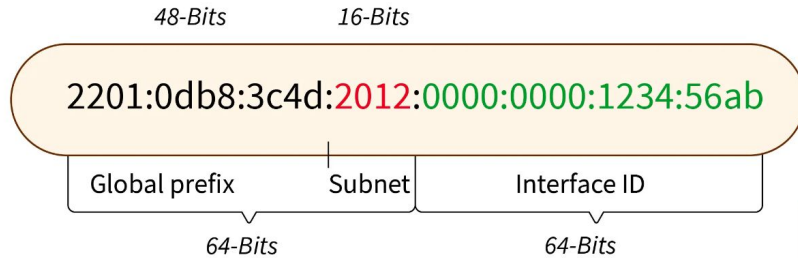
- ❖ You can **omit the leading zeros** of each 16-bit group, as follows:

3FFE:0:0:1:200:F8FF:FE75:50DF

- ❖ You can compress 16-bit groups of zeros to **double colons (::)** as shown in the following example, but only once per address:

3FFE::1:200:F8FF:FE75:50DF

► IPv6 Address Representation:

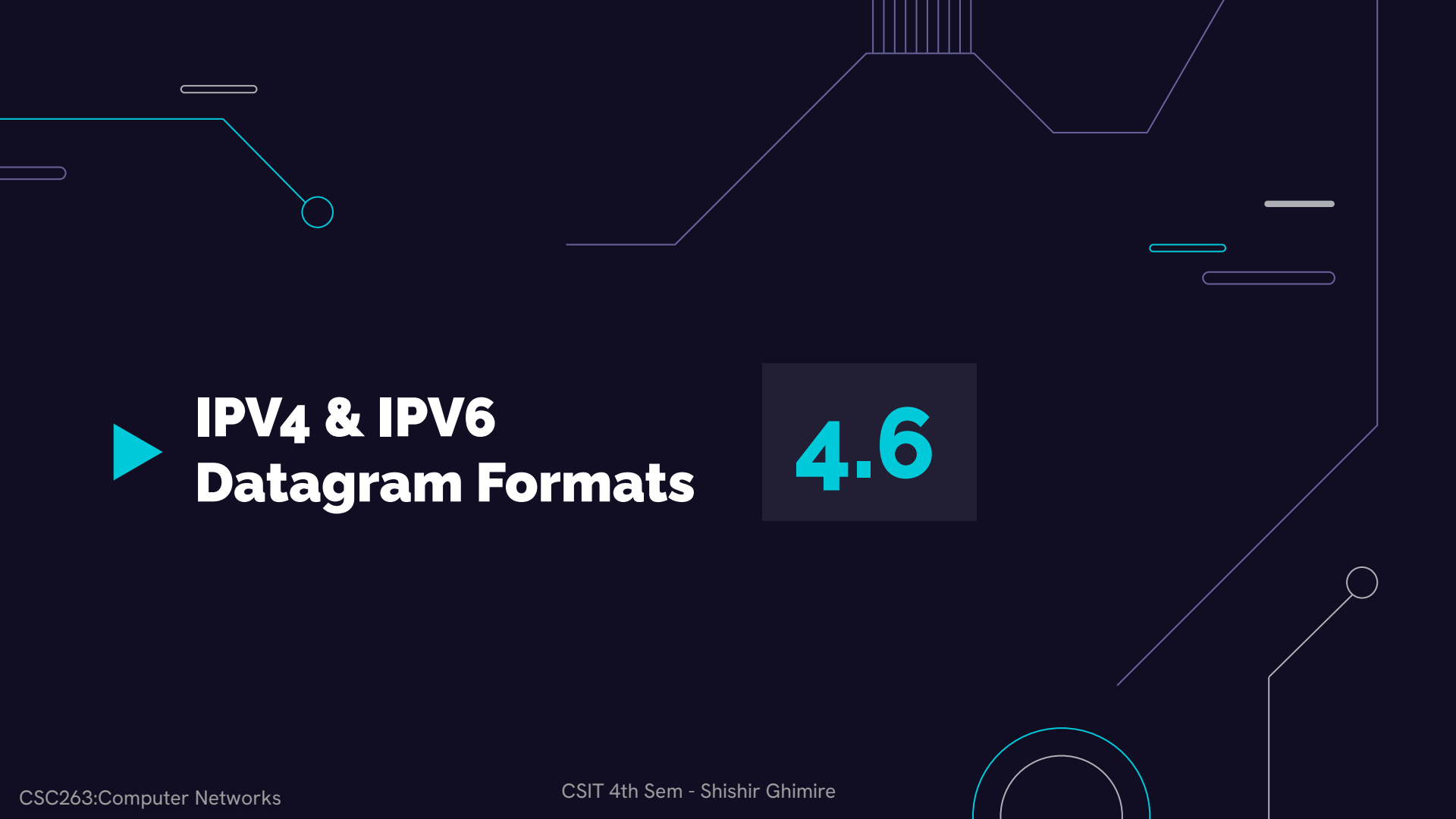


► Advantages of IPv6 Address

- ❖ **Efficient Routing** – With IPv6, routing becomes **more streamlined and hierarchical** while also shrinking the size of routing tables. In IPv6 networks, the source device manages fragmentation using a protocol for the detection of the path maximum transmission unit rather than a router.
- ❖ **Efficient Packet Processing** – The checksum **does not need** to be regenerated at each router hop because IPv6 does not include an IP-level checksum as IPv4 does.
- ❖ **Directed Data Flows** – Multicast is **supported** by IPv6 as opposed to broadcast. Network bandwidth can be **conserved by** using multicast to simultaneously send packet flows that use a lot of bandwidth to several destinations.
- ❖ **Security** – IPv6 is built using IPSec security, which offers data integrity, confidentiality, and authentication

► Disadvantages of IPv6 Address:

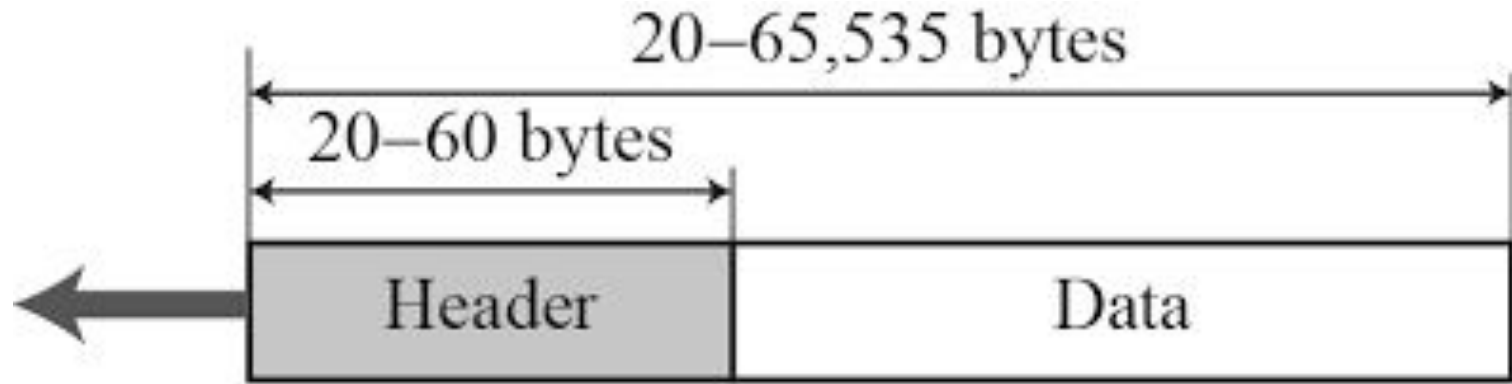
- ❖ The network infrastructure becomes **overworked when routing table entries** are needed for such a large number of networks.
- ❖ Older devices that **do not support** IPv6 must be manually set up using the Dual stack approach, adding extra effort for the network administrator.
- ❖ The globe is **slow to switch** to IPV6 and IPV4 is still widely used.



▶ **IPV4 & IPV6 Datagram Formats**

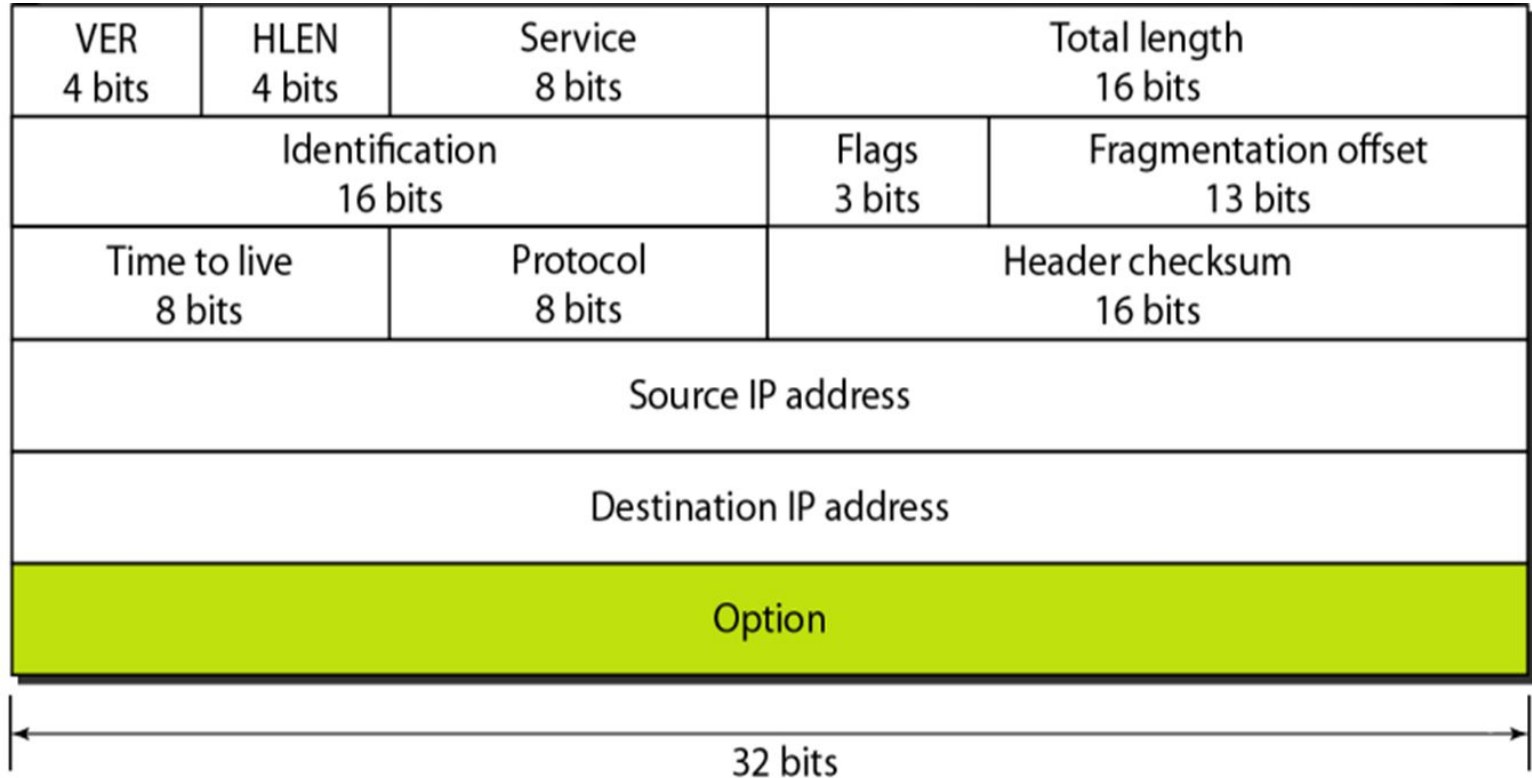
4.6

► IPv4 Datagram Format:



a. IP datagram

► IPv4 Header Format:



► IPv4 Addressing:

❖ Version:

- This 4-bit field defines the version of the IP protocol. Currently the version is 4.

❖ Header Length:

- This 4-bit field defines the total length of the datagram header in 4-byte words. This field is needed because the length of the header is variable (between 20 and 60 bytes).

❖ Service Type (8 bit):

- In the original design of IP header, this field was referred to as **type of service (TOS)**, which defines how the datagram should be handled.

❖ Total Length (16 bits):

- Total length of the datagram, measured in octets, including header and data.

❖ Identification (16 bits):

- A value assigned to aid in assembly of fragments.

IPV4 Addressing:

❖ Flags (3 bits):

- Various Control Flags are required if the size of packet is too big. (either to fragment or not).
- Bit 0: Reserved. Must be 0.
- Bit 1: (DF) 0 = May Fragment, 1 = Don't Fragment
- Bit 2: (MF) 0 = Last Fragment, 1 = More Fragments

❖ Time to Live (8 bits):

- Maximum time the datagram is allowed to exist in the system. Each router that handles the datagram decrements the TTL by 1.

❖ Fragment Offset (13 bits):

- Specifies the exact position of the fragment in the original IP Packet.

❖ Protocol:

- This 8-bit field defines the higher-level protocol that uses the services of the IP layer. An IP datagram can encapsulate data from several higher level protocols such as TCP, UDP, ICMP, and IGMP.

IPV4 Addressing:

- ❖ **Checksum:** It is used to detect error in the delivery of packet.
- ❖ **Source address:**
 - This 32-bit field defines the IP address of the source.
- ❖ **Destination address:**
 - This 32-bit field defines the IP address of the destination.
- ❖ **Options (variable length):**
 - An optional field that may include various options such as timestamp, record route, etc
- ❖ **An IPv4 packet has arrived with the first 8 bits as shown: 01000010 The receiver discards the packet. Why?**

Solution: There is an error in this packet. The 4 leftmost bits (0100) show the version, which is correct. The next 4 bits (0010) show an invalid header length ($2 \times 4 = 8$). The minimum number of bytes in the header must be 20. The packet has been corrupted in transmission.

IPV4 Header Analysis using Wireshark

Internet Protocol Version 4, Src: 192.168.82.147 (192.168.82.147), Dst: 192.243.232.2 (192.243.232.2)

- Version: 4
- Header Length: 20 bytes
- Differentiated Services Field: 0x00 (DSCP 0x00: Default; ECN: 0x00: Not-ECT (Not ECN-Capable Transport))
 - 0000 00.. = Differentiated Services Codepoint: Default (0x00)
 -00 = Explicit Congestion Notification: Not-ECT (Not ECN-Capable Transport) (0x00)
- Total Length: 1155
- Identification: 0x69de (27102)
- Flags: 0x02 (Don't Fragment)
 - 0... = Reserved bit: Not set
 - .1.. = Don't fragment: Set
 - ..0. = More fragments: Not set
- Fragment offset: 0
- Time to live: 128
- Protocol: TCP (6)
- Header checksum: 0xd064 [validation disabled]
 - [Good: False]
 - [Bad: False]
- Source: 192.168.82.147 (192.168.82.147)
- Destination: 192.243.232.2 (192.243.232.2)
 - [Source GeoIP: Unknown]
 - [Destination GeoIP: Unknown]

Transmission Control Protocol, Src Port: 57487 (57487), Dst Port: 80 (80), Seq: 1102, Ack: 883, Len: 1115

► IPv6 Datagram Format:

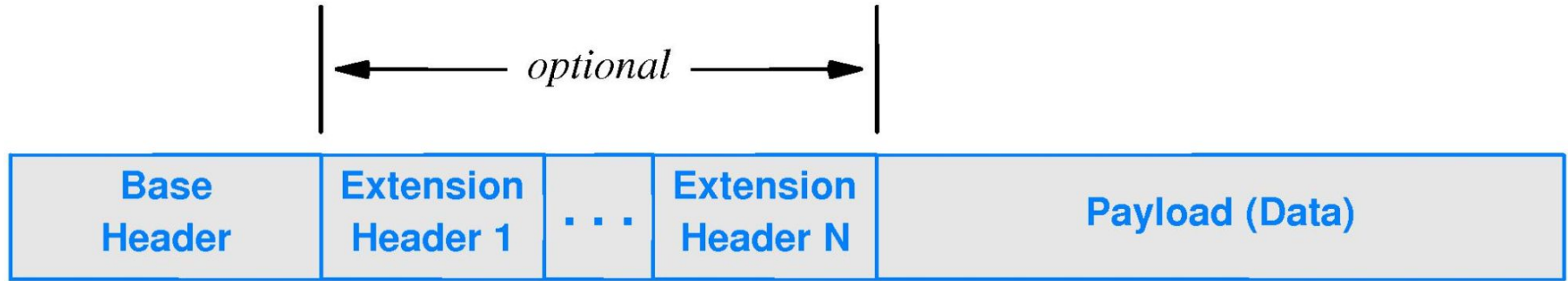


Figure 22.3 The general form of an IPv6 datagram.

► IPv6 Header Format:

Version (4)	Traffic Class (8)	Flow label (20)	
Payload length (16)		Next header (8)	Hop limit (8)
Source Address (128)			
Destination Address (128)			

► IPv6 Addressing:

- ❖ **Version (4-bits) :**

- Indicates version of Internet Protocol which contains bit sequence 0110.

- ❖ **Traffic Class (8-bits) :**

- The Traffic Class field indicates class or priority of IPv6 packet which is similar to Service Field in IPv4 packet. It helps routers to handle the traffic based on priority of the packet. If congestion occurs on router then packets with least priority will be discarded.

- ❖ **Flow Label (20-bits) :**

- Flow Label field is used by source to label the packets belonging to the same flow in order to request special handling by intermediate IPv6 routers, such as non-default quality of service or real time service.

► IPv6 Addressing:

❖ Payload Length (16-bits) :

- It is a 16-bit (unsigned integer) field, indicates total size of the payload which tells routers about amount of information a particular packet contains in its payload.

❖ Next Header (8-bits) :

- It identifies the type of the header protocol to which the data of the datagram will be delivered (UDP or TCP etc).

❖ Hop Limit (8-bits) :

- Hop Limit field is same as TTL in IPv4 packets.

❖ Source Address (128-bits) :

- Source Address is 128-bit IPv6 address of the original source of the packet.

❖ Destination Address (128-bits) :

- Destination Address field indicates the IPv6 address of the final destination.

► IPV6 vs IPV4 Datagram/Header Format:

IPv6 Header

Version	Traffic Class	Flow Label	
Payload Length		Next Header	Hop Limit
Source Address			
Destination Address			

IPv4 Header

Version	IHL	Type of Service	Total Length	
Identification			Flags	Fragment Offset
TTL	Protocol	Header Checksum		
Source Address				
Destination Address				
Options			Padding	

Legend

	Fields kept in IPv6
	Fields kept in IPv6, but name and position changed
	Fields not kept in IPv6
	Fields that are new in IPv6

► **Comparison of IPV4
and IPV6 Addressing:**

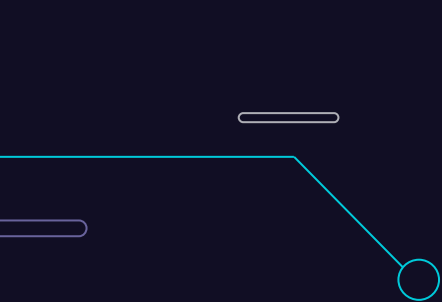
4.7

► **IPV4 vs IPV6**

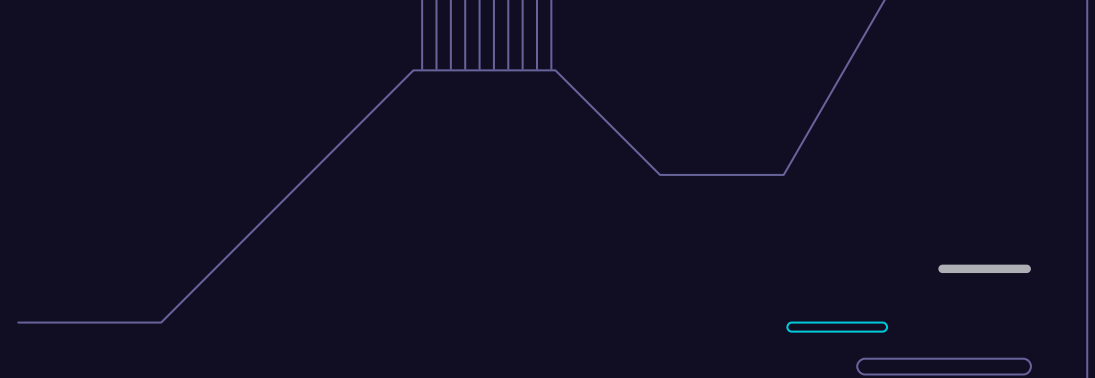
- ❖ IPv4 is 32-Bit IP address whereas IPv6 is a 128-Bit IP address.
- ❖ IPv4 is a numeric addressing method whereas IPv6 is an alphanumeric addressing method.
- ❖ IPv4 binary bits are separated by a dot(.) whereas IPv6 binary bits are separated by a colon(:).
- ❖ IPv4 offers 12 header fields whereas IPv6 offers 8 header fields.
- ❖ IPv4 supports broadcast whereas IPv6 doesn't support broadcast.
- ❖ IPv4 has checksum fields while IPv6 doesn't have checksum fields
- ❖ IPv4 supports VLSM (Virtual Length Subnet Mask) whereas IPv6 doesn't support VLSM.
- ❖ IPv4 uses ARP (Address Resolution Protocol) to map to MAC address whereas IPv6 uses NDP (Neighbour Discovery Protocol) to map to MAC address.

► IPV4 vs IPV6:

	Ipv4	Ipv6
Address length	IPv4 is a 32-bit address.	IPv6 is a 128-bit address.
Fields	IPv4 is a numeric address that consists of 4 fields which are separated by dot (.).	IPv6 is an alphanumeric address that consists of 8 fields, which are separated by colon.
Classes	IPv4 has 5 different classes of IP address that includes Class A, Class B, Class C, Class D, and Class E.	IPv6 does not contain classes of IP addresses.
Number of IP address	IPv4 has a limited number of IP addresses.	IPv6 has a large number of IP addresses.
VLSM	It supports VLSM (Virtual Length Subnet Mask). Here, VLSM means that Ipv4 converts IP addresses into a subnet of different sizes.	It does not support VLSM.
Address configuration	It supports manual and DHCP configuration.	It supports manual, DHCP, auto-configuration, and renumbering.
Address space	It generates 4 billion unique addresses	It generates 340 undecillion unique addresses.
End-to-end connection integrity	In IPv4, end-to-end connection integrity is unachievable.	In the case of IPv6, end-to-end connection integrity is achievable.
Security features	In IPv4, security depends on the application. This IP address is not developed in keeping the security feature in mind.	In IPv6, IPSEC is developed for security purposes.
Address representation	In IPv4, the IP address is represented in decimal.	In IPv6, the representation of the IP address in hexadecimal.
Fragmentation	Fragmentation is done by the senders and the forwarding routers.	Fragmentation is done by the senders only.
Packet flow identification	It does not provide any mechanism for packet flow identification.	It uses flow label field in the header for the packet flow identification.
Checksum field	The checksum field is available in IPv4.	The checksum field is not available in IPv6.
Transmission scheme	IPv4 is broadcasting.	On the other hand, IPv6 is multicasting, which provides efficient network operations.
Encryption and Authentication	It does not provide encryption and authentication.	It provides encryption and authentication.
Number of octets	It consists of 4 octets.	It consists of 8 fields, and each field contains 2 octets. Therefore, the total number of octets in IPv6 is 16.



► NATing



4.8

► NATing

- ❖ More people now use the internet than ever anticipated, leading to the situation where the number of unique IP addresses provided by the IPv4, 2^{32} is **not enough**.
- ❖ Network Address Translation(NAT) is one of the **solutions** to the problem because it requires only a **single IP address** to represent an **entire group of computers**.
- ❖ The number of IP addresses that are currently available is just insufficient considering the growth of the internet and the number of household and corporate networks.
- ❖ Network Address Translation(NAT) is a process of **assigning a unique public IP address** to represent an entire group of computers.
- ❖ In Network Address Translation, a network device, typically a **router or NAT firewall**—assigns a **public address** to one or more devices connected to a **private network**. Network address translation enables a single device to serve **as an agent** or intermediate between a **local, private network and the internet, a public network**. The basic objective of NAT is to **reduce the number** of public IP addresses in use for security and financial reasons.

► NATing

Devices
(PC/Tablet/SmartPhone)



Network Address Translation
(NAT)

Router



Private IP Address:
174.122.1.1

Public IP Address:
244.46.1.1

Internet



► NATing

- ❖ NAT enables devices on a **private network** to communicate with **external networks like the internet** by translating **private IP addresses** into a **single public IP address**.
- ❖ This process allows multiple devices to **share** a single IP address for external communication, **hiding** the private IP addresses from the public.

► NAT Inside & Outside Addresses:

- ❖ **Inside local address** – An IP address that is **assigned to a host** on the Inside (local) network. The address is probably not a IP address assigned by the service provider i.e., these are private IP address. This is the inside host seen from the inside network.
- ❖ **Inside global address** – IP address that **represents one or more inside local IP addresses** to the outside world. This is the inside host as seen from the outside network.
- ❖ **Outside local address** – This is the **actual IP address** of the destination host in the local network after translation.
- ❖ **Outside global address** – This is the outside host as seen from the outside network. It is the IP address of the **outside destination host** before translation.

► How NAT Works?

❖ Device Initiation:

- A device on the private network **initiates a connection** to a device on the internet. This device has a **private IP address** that is not routable on the public internet.

❖ Request Reception:

- The NAT device (e.g., a router or firewall) receives the request. It has a **public IP address** that is used for internet communication.

❖ IP and Port Translation:

- The NAT device translates the **private IP address** and **port number** of the internal device to its **public IP address** and a **unique port number**. This translation is recorded in a **NAT translation table** to keep track of active connections.

❖ Forwarding the Request:

- The NAT device forwards the translated request to the internet, using its public IP address. The destination server on the internet receives the request and responds to the NAT device public IP address.

► How NAT Works?

❖ Response Reception:

- The NAT device receives the response from the internet. It uses the **NAT translation table** to match the **public IP address and port number** back to the **original private IP address and port number** of the device that initiated the request.

❖ Response Delivery:

- The NAT device delivers the response to the **correct device** on the private network using the original private IP address and port number.

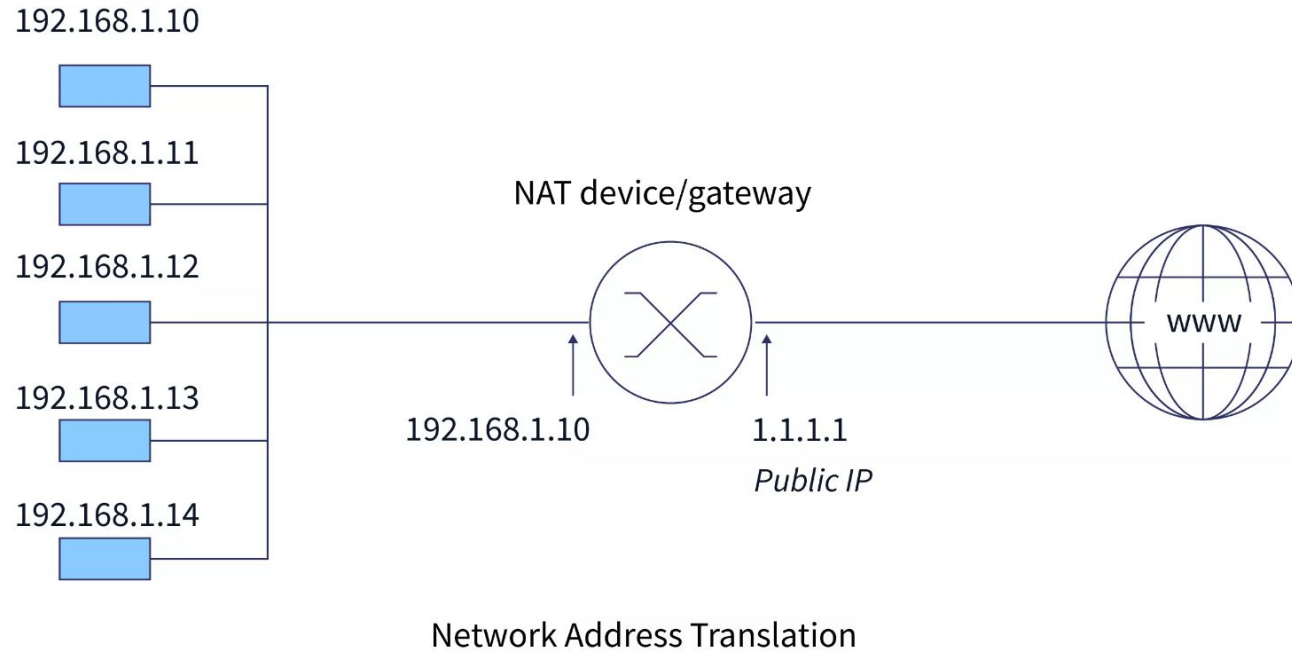
❖ Session Maintenance:

- The NAT device maintains the translation table for the **duration of the session**. It can handle multiple requests and responses **simultaneously** by keeping unique mappings for each session.

❖ Connection Termination:

- Once the session is complete, the **NAT entry is removed** from the translation table, freeing up resources for new connections.

► How NAT Works?



► How NAT Works (Example)?

❖ Initiating Communication:

- A device on the private network with an internal IP address (e.g., 192.168.1.10) wants to communicate with an external device, such as a web server on the internet.

❖ Address Translation:

- The NAT device (e.g., a router or firewall) translates the internal IP address (192.168.1.10) to the network's public IP address (1.1.1.1). This public IP is used for all external communications.

❖ Outgoing Request:

- The request from the internal device is sent to the internet using the public IP address (1.1.1.1). The original private IP is hidden from the outside world.

❖ Incoming Response:

- The web server responds to the public IP address (1.1.1.1). The NAT device receives the response and uses its records (NAT table) to map it back to the original internal IP address (192.168.1.10).

❖ Delivering the Response:

- The response is delivered to the appropriate device on the private network, allowing the internal device to receive the information it requested.

► Types of NAT:

❖ Static NAT

- **Mapping:** Each **private IP address** is mapped to a unique **public IP address**.
- **Usage:** Typically used for **hosting services** that require consistent access from the internet.
- **Cost:** Not practical for enterprises needing many IP addresses, as it **requires** purchasing a public IP for each device.

❖ Dynamic NAT

- **Mapping:** Private IP addresses are mapped to a **pool of** public IP addresses.
- **Usage:** Allows multiple private IPs to **share** a limited number of public IPs.
- **Limitation:** Only as many private IPs can access the internet as there are public IPs in the pool. If all are in use, additional requests are dropped.
- **Cost:** Requires buying several public IPs, which can be expensive but is more efficient than static NAT when the number of simultaneous connections is limited.

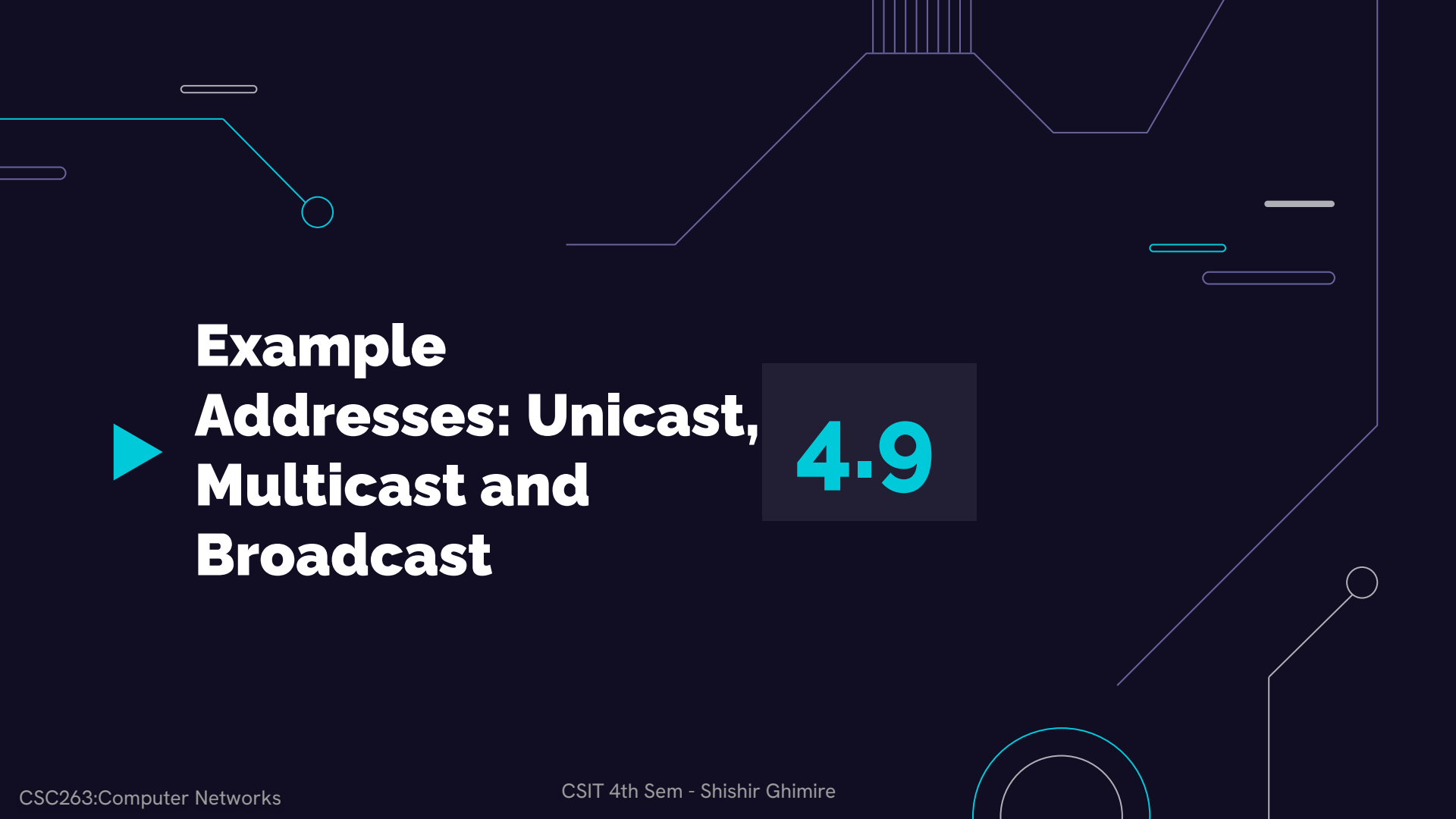
► Advantages and Disadvantages of NAT:

❖ Advantages of NAT

- NAT conserves legally registered IP addresses .
- It provides privacy as the device IP address, sending and receiving the traffic, will be hidden.
- Eliminates address renumbering when a network evolves.

❖ Disadvantage of NAT

- Translation results in switching path delays.
- Certain applications will not function while NAT is enabled.
- Complicates tunneling protocols such as IPsec.
- Also, router being a network layer device, should not tamper with port numbers(transport layer) but it has to do so because of NAT.



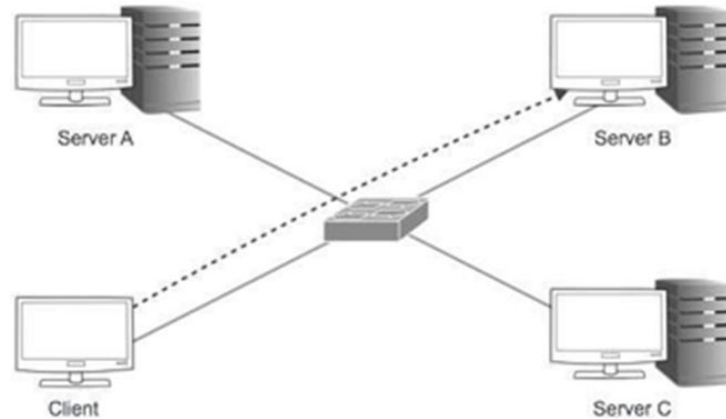
Example

► **Addresses: Unicast,
Multicast and
Broadcast**

4.9

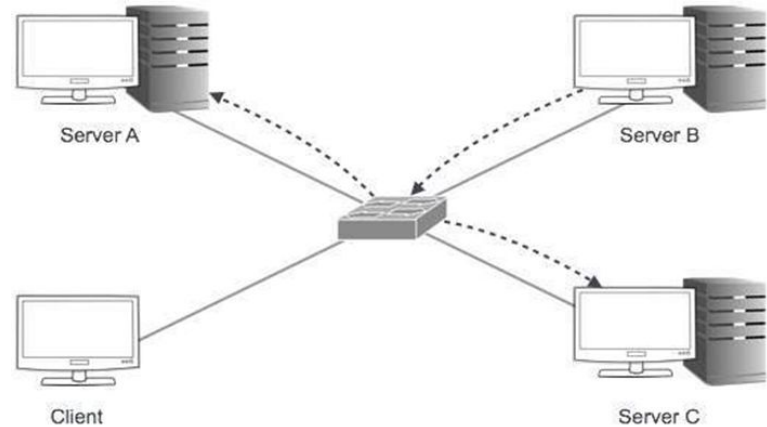
► Example Addresses:

- ❖ IPv4 supports three different types of addressing modes (Unicast, Multicast and Broadcast).
- ❖ Unicast:
 - In this mode, data is sent only to **one destined host**. The Destination Address field contains **32-bit IP address** of the destination host. Here the client sends data to the targeted server:



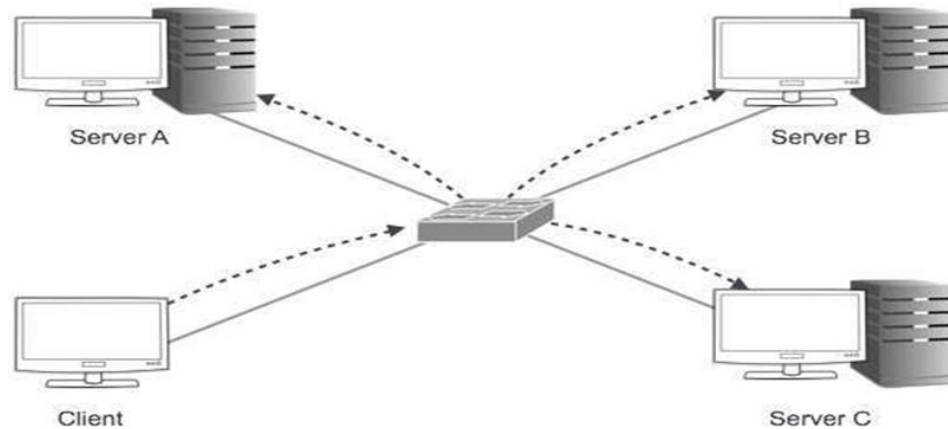
► Multicast:

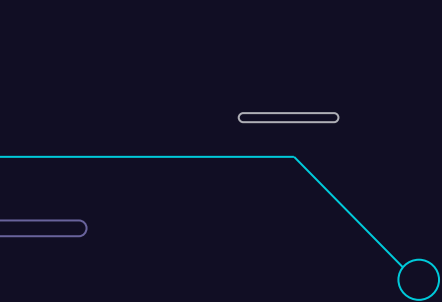
- ❖ This mode is a **mix of (unicast and broadcast)**, i.e. the packet sent is neither destined to a **single host nor all the hosts** on the segment. In this packet, the Destination Address contains a special address which **starts with 224.x.x.x** and can be entertained by more than one host.



► Broadcast:

- ❖ In this mode, the packet is addressed to **all the hosts** in a network segment. The Destination Address field contains a **special broadcast address**, i.e. 255.255.255.255. When a host sees this packet on the network, it is **bound to** process it. Here the client sends a packet, which is entertained by all the Servers:





► Routing



4.1
0

► Routing

- ❖ Process of **finding a path** from a source to every destination in the network .
- ❖ Suppose you want to **connect to China** from your desktop. **AND** Suppose you want to **connect to Antarctica** from your desktop.
 - What route should you take?
 - Does a shorter route exist?
 - What if a link along the route goes down?
 - What if you're on a mobile wireless link?
 - Routing deals with these types of issues.

► Routing

- ❖ A Routing is a process of **selecting path** along which the data can be transferred from source to the destination. Routing is performed by a special device known as a **router**.
- ❖ A Router works at the **network layer** in the OSI model and internet layer in TCP/IP model
- ❖ A router is a networking device that **forwards the packet** based on the information available in the packet header and **forwarding table**.
- ❖ The **routing algorithms** are used for routing the packets. The routing algorithm is nothing but a **software** responsible for deciding the **optimal path** through which packet can be transmitted.
- ❖ The routing protocols **use the metric** to determine the best path for the packet delivery.
- ❖ The **metric** is the standard of measurement such as hop count, bandwidth, delay, current load on the path, etc. used by the routing algorithm to determine the optimal path to the destination.
- ❖ The routing algorithm **initializes and maintains** the routing table for the process of path determination.

► Common Metric Values in Routing:

❖ Hop count:

- Hop count is defined as a metric that specifies the **number of passes** through internetworking devices such as a router, a packet must travel in a route to move from source to the destination.

❖ Delay:

- It is a time taken by the router to **process, queue and transmit** a datagram to an interface. The protocols use this metric to determine the delay values for all the links along the path end-to-end. The path having the **lowest delay value** will be considered as the best path.

❖ Bandwidth:

- The **capacity of the link** is known as a bandwidth of the link. The bandwidth is measured in terms of bits per second (**bps**). The protocol will determine the bandwidth capacity for all the links along the path, and the overall **higher bandwidth** will be considered as the best route.

► Common Metric Values in Routing:

◆ Load:

- Load refers to the degree to which the network resource such as a **router or network link is busy**. A Load can be calculated in a variety of ways such as CPU utilization, packets processed per second. If the traffic increases, then the load value will also be increased. The load value changes with respect to the change in the traffic.

◆ Reliability:

- Reliability is a metric factor may be composed of a fixed value. It depends on the network links, and its value is measured dynamically. Some networks go **down** more often than others. After network failure, some network links repaired more easily than other network links. Any reliability factor can be considered for the assignment of reliability ratings, which are generally numeric values assigned by the system administrator.

► 4.10.2 Types of Routing

- ❖ There are several types of routing some of them are listed below:
 - Static vs Dynamic
 - Unicast vs Multicast
 - Link State vs Distance Vector
 - Interior vs Exterior

► Static Routing (Non-Adaptive Routing)

- ❖ It is a technique in which the administrator **manually adds the routes** in a routing table.
- ❖ A Router can send the packets for the destination along the route **defined by the administrator**.
- ❖ In this technique, routing decisions are **not made** based on the condition or topology of the networks.
- ❖ **Advantages Of Static Routing**
 - **No Overhead:** It has No overhead on the **CPU usage** of the router. Therefore, the cheaper router can be used to obtain static routing.
 - **Bandwidth:** It has no bandwidth usage between the routers.
 - **Security:** It provides security as the system administrator is allowed **only to have control** over the routing to a particular network.
- ❖ **Disadvantages of Static Routing:**
 - For a large network, it becomes a **very difficult task** to add each route manually to the routing table.
 - The system administrator should have a **good knowledge of a topology** as he has to add each route manually.

► Dynamic Routing (Adaptive Routing)

- ❖ It is a technique in which a router **adds a new route** in the routing table for each packet in response to the **changes** in the condition or topology of the network.
- ❖ Dynamic protocols are used to **discover** the new routes to reach the destination.
- ❖ In Dynamic Routing, **RIP** (Routing Information Protocol) and **OSPF** (Open Shortest Path First) are the **protocols** used to discover the new routes.
- ❖ If any route goes down, then the **automatic adjustment** will be made to reach the destination.
- ❖ The Dynamic protocol should have the following features:
 - All the routers must have the **same dynamic routing protocol** in order to exchange the routes.
 - If the router **discovers any change** in the condition or topology, then router **broadcast** this information to all other routers.

► Dynamic Routing (Adaptive Routing)

❖ Advantages of Dynamic Routing:

- It is easier to configure.
- It is **more effective** in selecting the best route in response to the changes in the condition or topology.

❖ Disadvantages of Dynamic Routing:

- It is **more expensive** in terms of CPU and bandwidth usage.
- It is **less secure** as compared to default and static routing.

► Unicast vs Multicast

Feature	Unicast	Multicast
Recipients	One sender to one recipient.	One sender to multiple recipients.
Bandwidth Usage	Consumes separate bandwidth for each recipient.	Shares a single copy of data to reduce bandwidth usage.
Efficiency	Straightforward but may lead to inefficiencies for multiple recipients.	More efficient for delivering data to multiple recipients simultaneously.
Group Management	Requires individual connections for each recipient.	Efficient group management; recipients join a multicast group.
Applications	Common in one-on-one communication.	Suited for one-to-many scenarios.

► Distance Vector Routing:

- ❖ It is a dynamic routing algorithm in which each router computes distance **between** itself and each possible destination i.e immediate neighbours.
- ❖ The router **shares its knowledge** about the whole network to its neighbours and accordingly **updates tables** based on its neighbours.
- ❖ It makes use of **Bellman Ford Algorithm** for making routing tables.
- ❖ Each router maintains a distance table known as **Vector**.
- ❖ **RIP** (Routing Information Protocol) and **IGRP** (Internal Gateway Routing Protocol) are some examples of distance vector routing algorithms

► Distance Vector Routing (Working Methodology):

❖ Step1:

- Each routers prepares its routing tables.
- Each router has information
 - About neighboring routers.
 - Distance to them
 - Destination routers

❖ Step 2:

- Router shares distance vector with neighbor router and update routing table.
- This process is repeated **n-1 times** to finally make routing tables stable(converges) where n is number of router.

► Distance Vector Routing:

Few **key points** about the distance vector routing protocol:

◆ Network Information:

- Every node in the network should have information about its neighboring node. Each node in the network is **designed to share information** with all the nodes in the network.

◆ Routing Pattern:

- In DVR the data shared by the nodes are transmitted only to that **node that is linked** directly to **one or more nodes** in the network.

◆ Data sharing:

- The nodes share the information with the neighboring node from **time to time** as there is a change in network topology.

► Distance Vector Routing ► Bellman-Ford Algorithm:

❖ The **Bellman-Ford algorithm** is defined as:

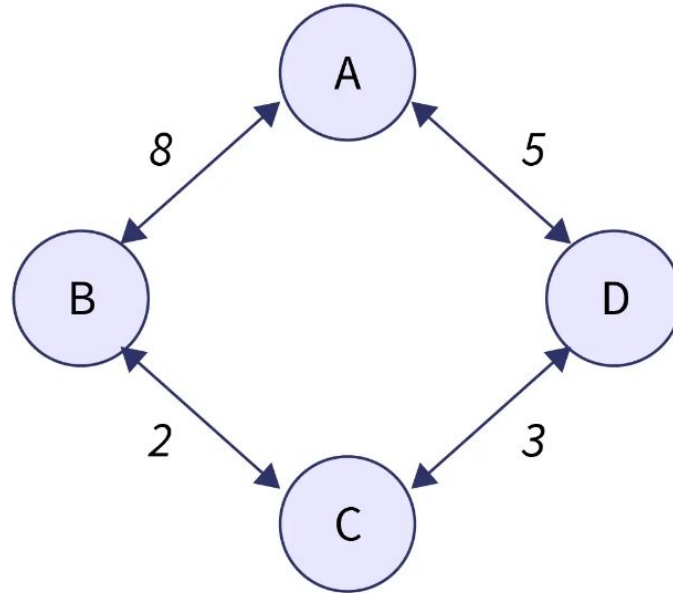
$$d_x(y) = \min_v \{c(x, v) + d_v(y)\}$$

❖ where,

- $d_x(y)$ = The least distance from x to y
- $c(x, v)$ = Node x's cost from each of its neighbor v
- $d_v(y)$ = Distance to each node from initial node
- min_v = selecting shortest distance

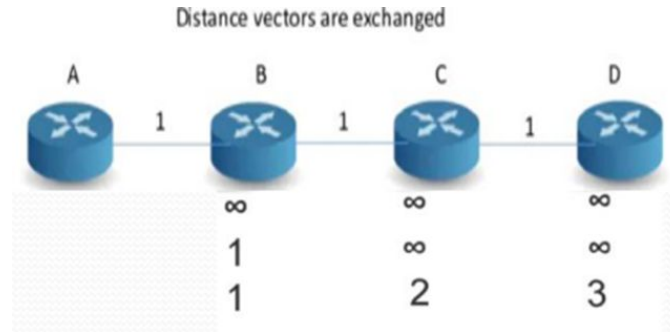
► Distance Vector Routing Example:

► Distance Vector Routing Example:



► Problem associated with DVR:

- ❖ Count to infinity problem in DVR.

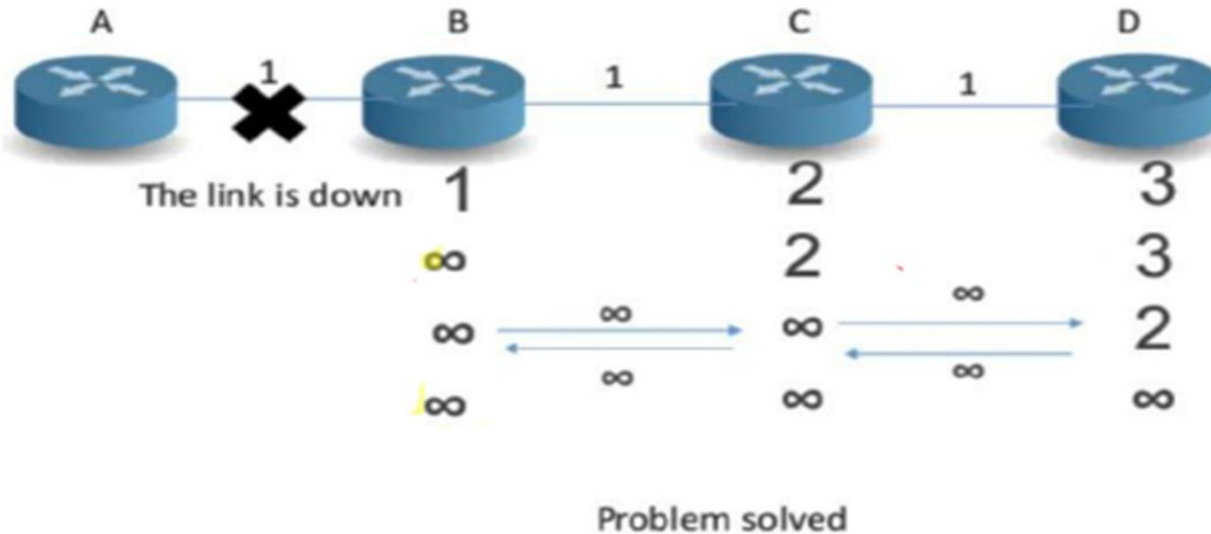


- ❖ Suppose there is link **down** between **A and B** then their occur **count to infinity** problem.
 - Count to infinity problem appears because **router only** exchanges **distance vector** and **not the next hop**.
- ❖ These problem can be solved using **Split horizon** and **poison reverse algorithm**.

► Problem associated with DVR:

- ❖ Split horizon" refers to a technique used in computer networking, particularly in the context of routing protocols, to prevent **routing loops**.
 - **Basic Split Horizon:** In this approach, a router simply **does not advertise** routes back through the interface from which it learned them. This is a simple and effective way to prevent routing loops.
 - **Split Horizon with Poison Reverse:** In this variation, the router advertises the route **back to the neighbour**, but with an infinite metric (infinite cost). This signals to the neighbor that the route is **unreachable**, preventing the neighbor from using that route in its routing decisions.

► Problem associated with DVR:



► Link State Routing:

- ❖ In LSR, every routers shares its routing table with every other router **using flooding**.
- ❖ Uses **Global Knowledge**
- ❖ After receiving routing tables from all routers **Dijkstra's algorithm** is used to find shortest distance.
- ❖ This will happen **every time a latest routing table** is received from routers.
- ❖ **Problems:**
 - Heavy traffic
 - Due to flooding and also each router has to maintain a **record** of the incoming **packets** and their **sequence numbers**.

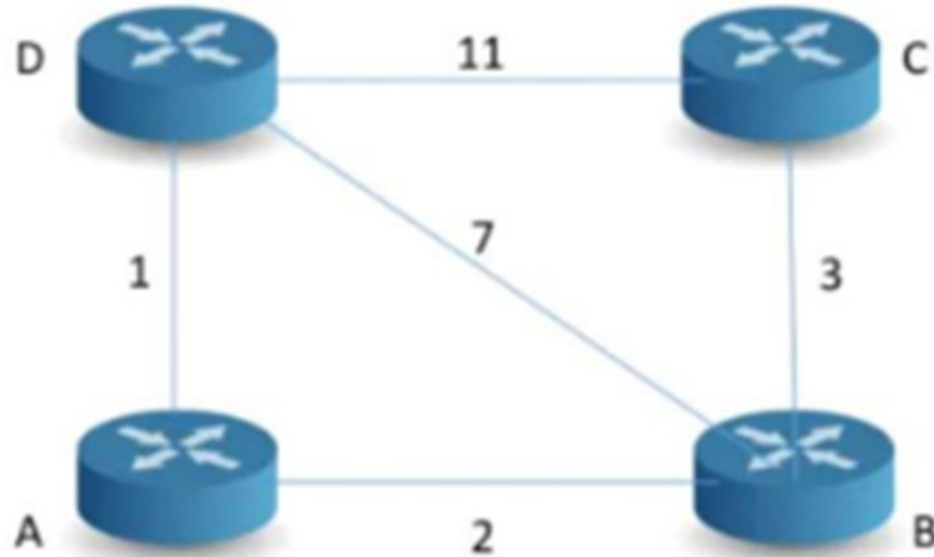
► Important Points Related to the Link State Routing Algorithm:

- ❖ The link state routing algorithm exchanges information **only when there is** a change in the connection.
- ❖ It requires **large memory** as it maintains a routing database.
- ❖ It requires the computation of the shortest path, which is an **overhead** for the CPU.
- ❖ The information of each router **needs to be transmitted** all over the network.

► Link State Routing:

- ❖ Step 1: **Finding Distance to Neighbouring Routers**
 - Begin by determining the distance to each neighbouring router from the source router.
 - For each neighbouring router, calculate the tentative distance based on the **link cost**.
- ❖ Step 2: **Prepare Link-State Table**
 - Construct a **Link-State Table** to store information about the network topology.
 - Include entries for each router, detailing its neighbours and the associated link costs.
- ❖ Step 3: **Apply Flooding and Dissemination of Link-State Table**
 - Utilize the **flooding technique** to disseminate the Link-State Table to other routers in the network.
 - Broadcast the Link-State Table to ensure that every router receives the most up-to-date information about the network's state.
- ❖ Step 4: **Apply Dijkstra's Algorithm for Shortest Path Finding and create final routing table.**

► Link State Routing:



► DVR vs LSR:

Feature	Distance Vector Routing	Link State Routing
Information Exchange	Summarized routing information exchanged with neighbors.	Detailed information about link states exchanged with the entire network.
Routing Table Updates	Periodic updates sent to neighbors regardless of changes.	Updates triggered by changes, and routers flood the network with current information.
Routing Table Size	Smaller routing tables with summarized information.	Larger routing tables with detailed information about the entire network.
Convergence Time	Slower convergence due to periodic updates and potential count-to-infinity problem.	Faster convergence as updates are triggered by changes, providing quicker adaptation.
Routing Algorithm	Bellman-Ford algorithm is commonly used.	Dijkstra's algorithm is commonly used for calculating shortest paths.
Overhead	Lower overhead as updates contain less detailed information.	Higher overhead due to detailed link state information being flooded.
Scalability	Less scalable for larger networks.	More scalable for larger and complex networks.
Routing Table Calculation	Calculations based on local information and information from neighbors.	Independently calculates shortest paths using detailed network topology information.
Example Protocols	RIP (Routing Information Protocol) is a distance vector protocol.	OSPF (Open Shortest Path First) is a link state protocol.
Handling Changes	Slower adaptation to changes in the network.	Faster adaptation to changes, immediate response to network modifications.

► Interior vs Exterior:

Feature	Interior Gateway Protocols (IGPs)	Exterior Gateway Protocols (EGPs)
Scope	Operate within a single autonomous system (AS).	Facilitate routing between different autonomous systems (ASes).
Topology Information	Have detailed information about the internal topology of an AS.	Focus on high-level routing information and lack detailed knowledge of internal AS structures.
Examples	Examples include RIP, OSPF, EIGRP.	Examples include BGP (Border Gateway Protocol).
Admin Control	Network administrators have full control over the configuration within their AS.	Involves negotiations and agreements between different ASes, with less direct control at the administrative level.
Routing Decision Factors	Routing decisions based on metrics like hop count, bandwidth, or delay within the AS.	Routing decisions influenced by policies, business relationships, and metrics related to reaching networks across different ASes.

▶ 4.10.3 Path Computation Algorithms: Bellman Ford, Dijkstra's

- ❖ Path computation algorithms are algorithms **designed to find** the optimal or feasible paths between two points in a network or system.
- ❖ These points can represent nodes in a computer network.
- ❖ Both the **Bellman-Ford algorithm** and **Dijkstra's algorithm** are path computation algorithms used in computer networks to find the shortest paths from a source node to all other nodes in a graph.
- ❖ Both algorithm are **single-source algorithms** used for finding the shortest paths in a graph.

► Bellman Ford Algorithm:

- ❖ The Bellman-Ford algorithm is a way to find **single source shortest paths** in a graph with negative edge weights (but no negative cycles).
- ❖ The first for loop relaxes each of the edges in the graph **$n - 1$ times**.
- ❖ We claim that after $n - 1$ iterations, the distances are **guaranteed** to be correct.
- ❖ The second for loop in this algorithm also detects negative cycles.

► Bellman Ford Algorithm:

Following are the detailed steps.

- ❖ This step **initializes distances** from the source to all vertices as **infinite** and distance to the **source** itself as 0. Create an array $\text{dist}[]$ of size $|V|$ with all values as infinite except $\text{dist}[\text{src}]$ where src is source vertex.
- ❖ This step calculates shortest distances. Do following **$|V|-1$ times** where $|V|$ is the number of vertices in given graph.
- ❖ Do following for **each edge** u to v .
 - IF $\text{dist}[v] > \text{dist}[u] + \text{weight of edge } uv$, then update $\text{dist}[v]$ as $\text{dist}[v] = \text{dist}[u] + \text{weight of edge } uv$

► Bellman Ford Algorithm:

- ❖ This step reports if there is a **negative weight cycle** in graph.
- ❖ Do following for each edge u to v
 - If $\text{dist}[v] > \text{dist}[u] + \text{weight of edge } (u,v)$, then "Graph contains negative weight cycle"
- ❖ **Note:**
 - **Time complexity:** $O(|V| * |E|)$
 - Can handle **negative edge weights** (except for negative-weight cycles)
 - Detects negative-weight **cycles**
 - Often used in routing algorithms and network optimization

► Bellman Ford Algorithm:

❖ References:

- https://www.geeksforgeeks.org/bellman-ford-algorithm-simple-implementation/?ref=oin_asr2
- <https://www.javatpoint.com/bellman-ford-algorithm>
- <https://www.youtube.com/watch?v=FtN3BYH2Zes>

► Dijkstra's Algorithm:

❖ 1.Initialization:

- Create two sets:
 - **Unvisited set:** Contains all vertices in the graph.
 - **Visited set:** Initially empty.
- Create a **distance array** dist of size V (number of vertices) and initialize all distances **to infinity** except for the source vertex, which is set to 0.
- Create a previous array of size V to **store the predecessor** of each vertex in the shortest path tree.

❖ 2.Find the Closest Unvisited Vertex:

- Choose the vertex **from the unvisited set** with the **smallest distance value** in the dist array.
- Remove this vertex from the unvisited set and **add it to the visited set.**

► Dijkstra's Algorithm:

❖ Update Distances for Neighbours:

- For each neighbour v of the chosen vertex:
 - **Calculate the tentative distance:** $\text{tentative_dist} = \text{dist}[\text{chosen_vertex}] + \text{weight}(\text{chosen_vertex}, v)$
 - If $\text{tentative_dist} < \text{dist}[v]$:
 - Update $\text{dist}[v]$ to tentative_dist
 - Update $\text{previous}[v]$ to chosen_vertex

❖ Repeat Steps 2 and 3 Until the Unvisited Set is Empty:

- Continue choosing the closest unvisited vertex and updating distances for its neighbors **until the unvisited set is empty.**

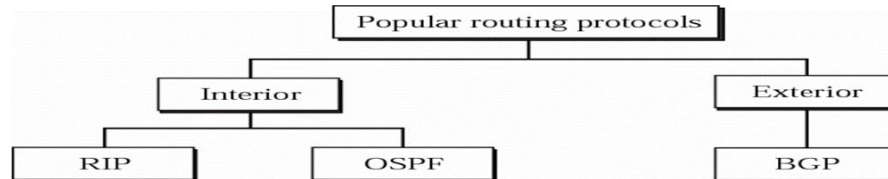
► Dijkstra's Algorithm:

❖ References:

- <https://www.geeksforgeeks.org/introduction-to-dijkstras-shortest-path-algorithm/>
- <https://www.youtube.com/watch?v=XB4MlexjvY0>

► 4.10.4 Routing Protocols: RIP, OSPF & BGP

- ❖ To determine the **path taken by a datagram** between source and destination.
- ❖ An autonomous-system (AS) is a collection of routers under the **same administrative control**. OR An autonomous system is a collection of IP networks and routers under the **control of a single organization** that presents a common routing policy to the internet.
- ❖ In AS, all routers run the **same routing protocol** among themselves.
- ❖ Most common intra-AS routing protocols:
 - Routing-information Protocol (RIP)
 - Open Shortest Path First (OSPF)
- ❖ Intra-AS routing protocols are also known as **interior gateway protocols**.



► Interior Routing Protocols:

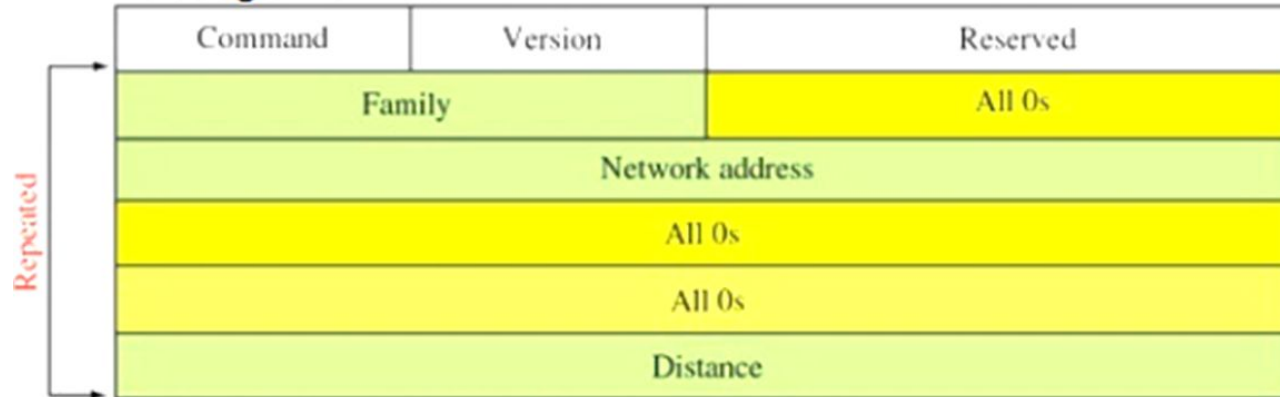
- ❖ An interior gateway protocol (IGP) is a type of protocol used for exchanging routing information between gateways (commonly routers) **within** an autonomous system (for example, a system of corporate local area networks).
- ❖ Interior gateway protocols can be divided into **two categories**:
 - Distance-vector routing protocols
 - Link-state routing protocols.
- ❖ Specific examples of IGP protocols include Open Shortest Path First (**OSPF**), Routing Information Protocol (**RIP**) and Intermediate System to Intermediate System (**IS-IS**).

► RIP (Routing Information Protocol):

- ❖ RIP is a **standardized** Distance Vector Routing Protocol, designed for use on **smaller networks**.
- ❖ RIP was one of the **first true Distance Vector Routing protocols**, and is supported on a wide variety of systems.
- ❖ RIP adheres to the following Distance Vector characteristics:
 - RIP sends out **periodic routing updates** (every 30 seconds)
 - RIP sends out the full routing table **every periodic update**.
 - RIP uses a form of distance as its **metric** (in this case, hop count)
 - RIP uses the **Bellman-Ford Distance Vector algorithm** to determine the best “path” to a particular destination .

► RIP (Routing Information Protocol):

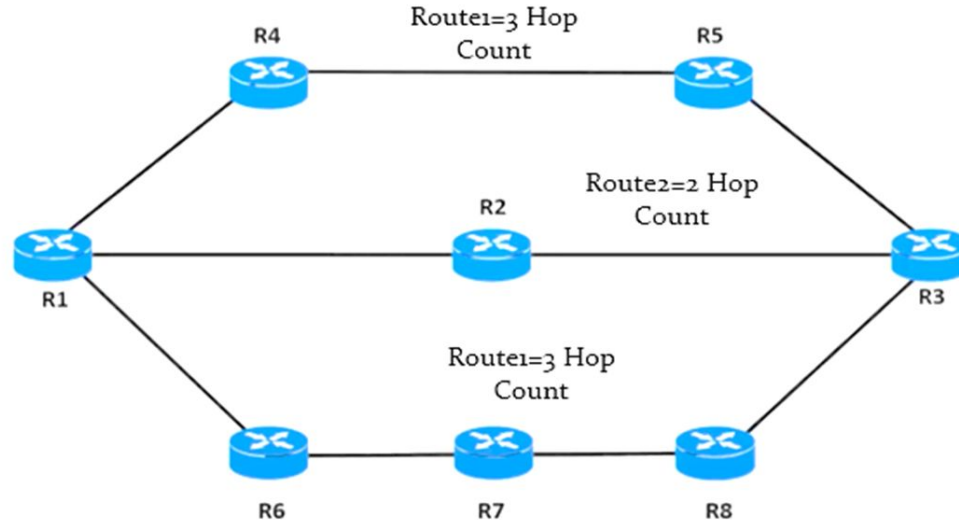
RIP Message Format



- Command : request (1) or response (2)
- Version
- Family : For TCP/IP the value is 2
- Address : destination network address
- Distance : defining the hop count from the advertising router to the destination network

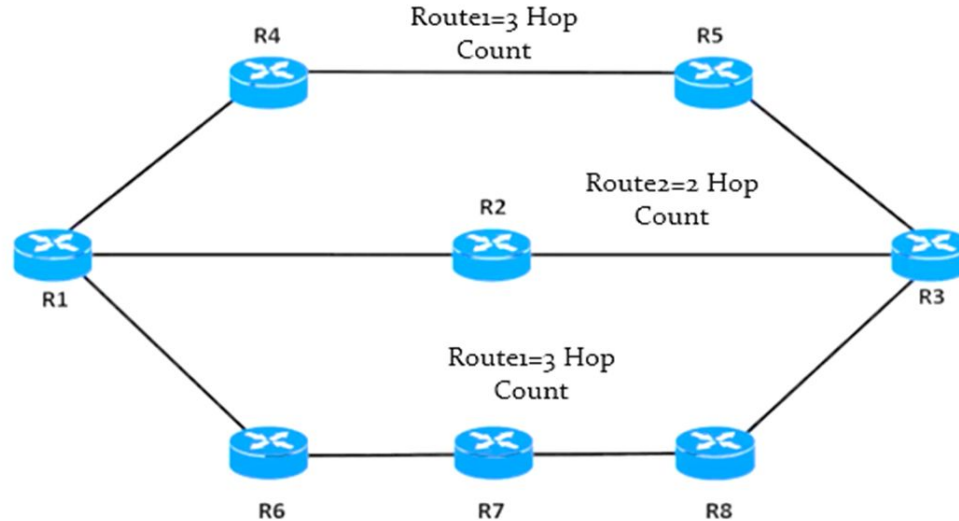
► RIP (Routing Information Protocol):

◆ Q: Which Path does RIP protocol chooses?



► RIP (Routing Information Protocol):

- ❖ If RIP is implemented in this network then **RIP chooses path 2** because route 2 has less number of Hop Count.

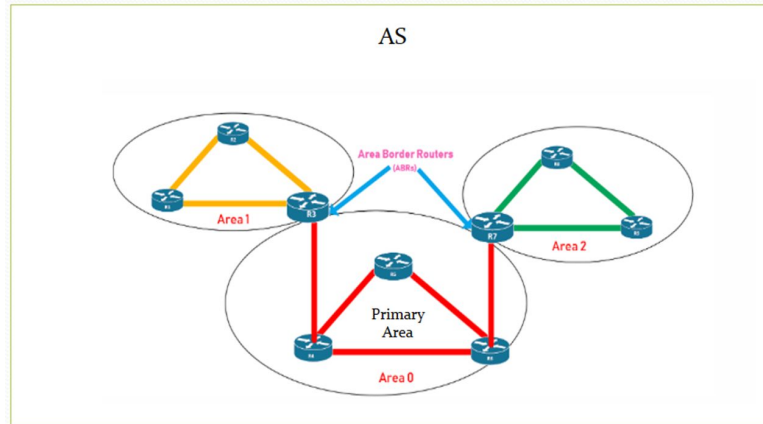


► OSPF(Open Shortest Path First):

- ❖ It stands for “**Open Shortest Path First**”.
 - Open Standard Protocol
 - Know By RFC 2328 & RFC 2740.
- ❖ OSPF is a **link-state** routing protocol & Classless Protocol.
- ❖ OSPF uses **Dijkstra's** shortest path first (SPF) algorithm to calculate the shortest path to a destination network.
- ❖ OSPF routers exchange information about the network using **link-state advertisements** (LSAs).
- ❖ Each router sends out LSAs to inform other routers of **its links and their states**.

► OSPF Area:

- ❖ OSPF divides autonomous system into **areas** which is a collection of networks, hosts and routers.
- ❖ Router **inside the area** flood the area about routing information.
- ❖ Area Border Router (ABR's) router **summarize the information** about the area and send it to other areas.
- ❖ All the areas are most **connected to backbone** of network or Primary Area.



► OSPF Tables:

❖ 1- Neighbour Table :

- This table lists all the **OSPF routers** that are directly connected to the **local router**.
- It includes information such as:-
 - the neighbour's IP address
 - router ID
 - the state of the adjacency.

❖ 2- Topology Table:

- This table contains the **complete topology of the network** used to calculate the shortest path to a destination network.
- It includes information about:-
 - all the routers, networks, and links in the network, along with their associated costs

► OSPF Tables:

❖ 3- Routing Table:

- This table contains the **best path** to each destination network in the network. It is derived from the topology table.
- It includes information such as:-
 - the next hop router
 - the cost of the path
 - the outgoing interface.

❖ 4- Link-state database:

- This database contains all the **LSAs(Link State Advertisements)** that have been received from other OSPF routers.
- It includes information about: -
 - Links, Routers, Networks in the Network.
 - The link-state database is used to build the topology table.

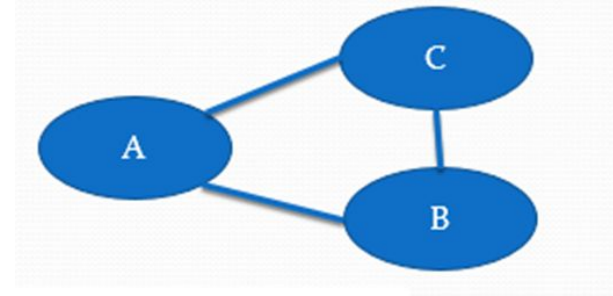
► OSPF Tables:

❖ 5- Forwarding table:-

- This table is **used by the router** to forward packets to their destination.
- It is **derived** from the routing table.
- It includes information such as:-
 - Destination network.
 - The Next Hop Router.
 - The Outgoing Interface.

► OSPF Area:

- ❖ Let's consider a simple network with **three routers**: Router A, Router B, and Router C and connected to each other with **unit cost**.



- ❖ Here for **Router A**,

1. Neighbor Table:

Neighbor's IP Address	Router ID	State of Adjacency
B's IP	2.2.2.2	Full
C's IP	3.3.3.3	Full

► OSPF Area:

2. Topology Table:

Router ID	Network/Link	Cost
1.1.1.1	A's network	0
2.2.2.2	B's network	1
3.3.3.3	C's network	1

3. Routing Table:

Destination Network	Next Hop Router	Cost	Outgoing Interface
A's network	-	0	Local Interface
B's network	2.2.2.2	1	Interface to B
C's network	3.3.3.3	1	Interface to C

4. Link-State Database:

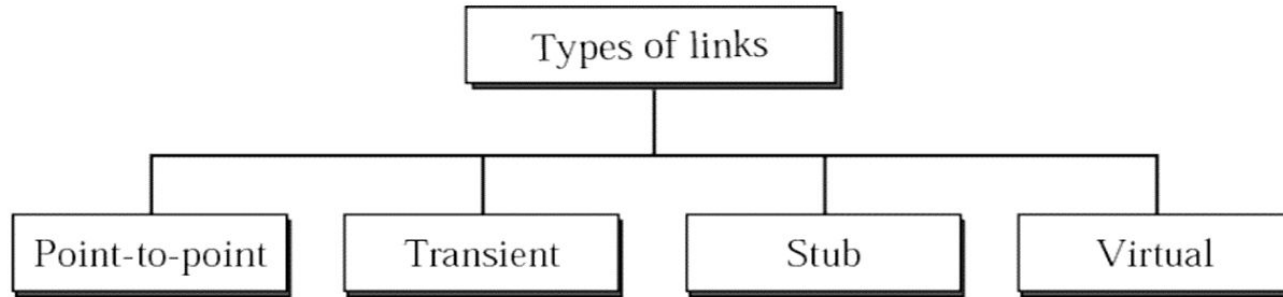
LSA Type	Origin Router	Link Information
Type 1	1.1.1.1	Router A's links and connected networks
Type 1	2.2.2.2	Router B's links and connected networks
Type 1	3.3.3.3	Router C's links and connected networks
Type 2	1.1.1.1	Summary LSA for networks outside the OSPF Autonomous System

5. Forwarding Table:

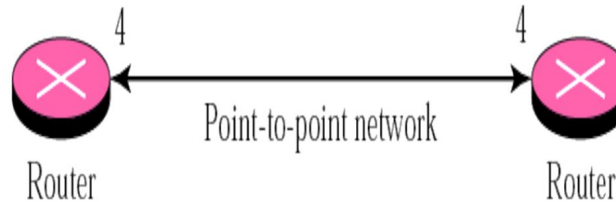
Destination Network	Next Hop Router	Outgoing Interface
A's network	-	Local Interface
B's network	2.2.2.2	Interface to B
C's network	3.3.3.3	Interface to C

► Types of Links:

- ❖ In OSPF, a **connection** is called a link.



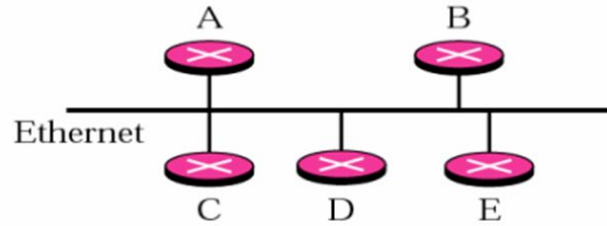
- ❖ Point to Point:
 - Connect two routers without any other host or router in between.



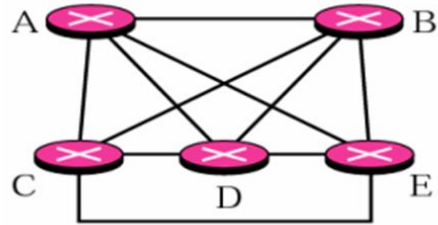
► Transient Link:

- ❖ A network with several routers attached to it.
- ❖ Data can enter through any of the routers and leave through any router.
- ❖ Unrealistic:
 - When all the routers are connected to each other, it is known as an unrealistic transient link.
- ❖ Realistic:
 - When some designated router exists in a network then it is known as a realistic transient link. Here designated router is a router to which all the routers are connected.
 - All the packets sent by the routers will be passed through the designated router.

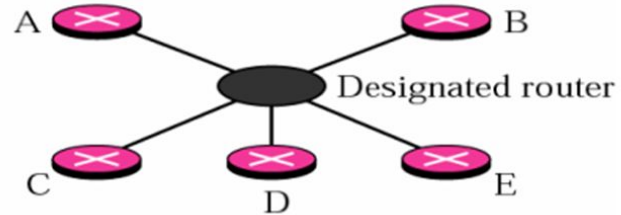
► Transient Link:



a. Transient network



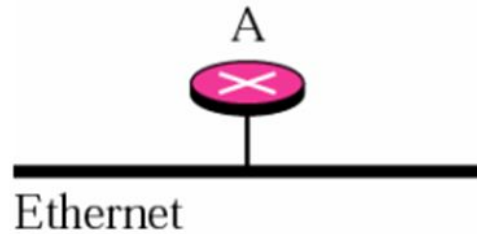
b. Unrealistic representation



c. Realistic representation

► Stub Link:

- ❖ A network that is connected to only one router.
- ❖ Data packet enter and leave through this only one router.



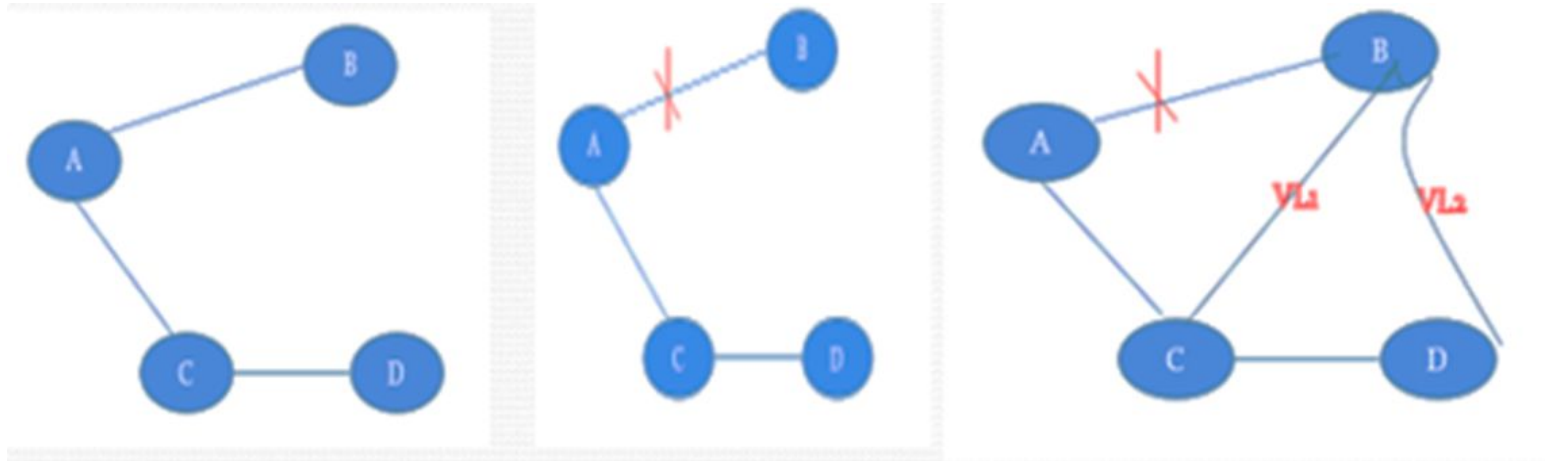
a. Stub network



b. Representation

► Virtual Link:

- ❖ When the link between two routers is broken, the administrator may create a virtual path between them using a longer path and may go through several routers.



► OSPF packets:

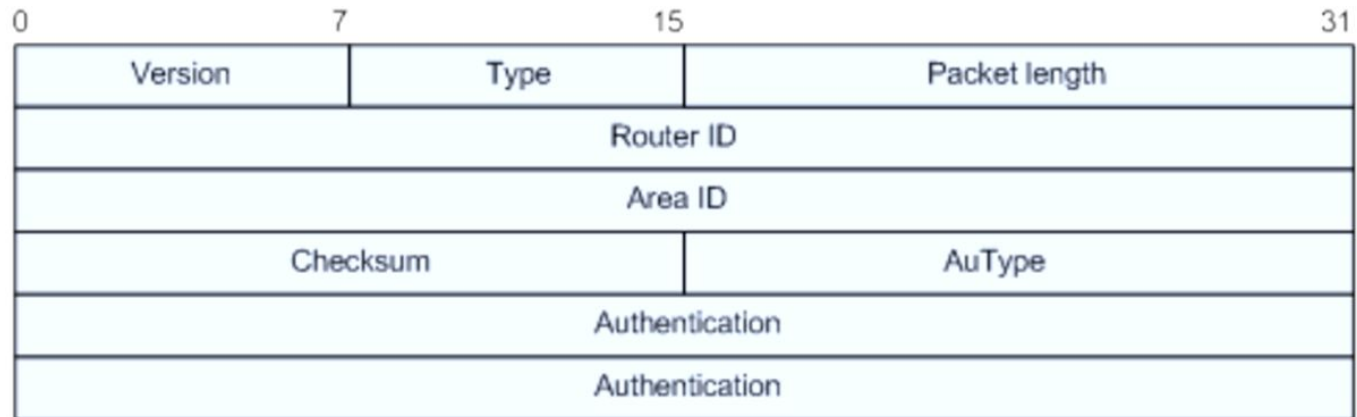
- ❖ **Hello**
 - It helps to form neighbor discovery.
- ❖ **Database Description(DBD)**
 - It helps to check the **LSDB (Link State Database)** between the neighbors are same or not.
- ❖ **Link state Request (LSR)**
 - Request the specific link state information from neighbor.
- ❖ **Link State Update(LSU)**
 - Reply to the link state request.
- ❖ **Link State ACK(LSAck)**
 - OSPF is **reliable protocol**, so it need to be acknowledged.

► OSPF Packets Formats:

- ❖ OSPF packets are **directly encapsulated** into IP packets. OSPF uses the IP protocol **number 89** (IP Protocol number is used in Field of IPV4 header).



❖ OSPF Packet Header



► Major Fields of OSPF packet header:

- ❖ **Version**—OSPF version number, which is 2 for OSPFv2.
- ❖ **Type**—OSPF packet type from 1 to 5, **corresponding to** HELLO, DD, LSR, LSU, and LSAck, respectively.
- ❖ **Packet length**—Total length of the **OSPF packet** in bytes, including the header.
- ❖ **Router ID**—ID of the advertising router.
- ❖ **Area ID**—ID of the area where the advertising router resides.
- ❖ **Checksum**—Checksum of the message.
- ❖ **Authentication Type**—Authentication type, ranging from 0 to 2, **corresponding to** non-authentication, simple (plaintext) authentication, and MD5 authentication, respectively.
- ❖ **Authentication**—Information determined by authentication type.
 - It is **not defined** for authentication type 0.
 - It is **defined as** password information for authentication type 1, and defined as Key ID, MD5 authentication data length, and **sequence number** for authentication type 2.

► Advantages OF OSPF:

- ❖ **Fast Convergence:** OSPF algorithm is very **efficient and can converge** quickly, meaning that network changes can be detected and responded quickly.
- ❖ **Scalability:** Used a hierarchical design. It supports areas, which can be used to **divide a large network into smaller**. This helps to reduce the size of the routing table and limit the propagation of routing information.
- ❖ **Load Balancing:** OSPF supports **multiple paths** to a destination network.
- ❖ **Security:** Supports authentication, write with line alignment.

Disadvantages:

- ❖ **Resource requirements:** OSPF requires **more resources** than some other routing protocols, particularly in terms of CPU and memory. This can be a disadvantage in smaller or less powerful networks. Can **reduce this** by Using Different Types Of Areas.

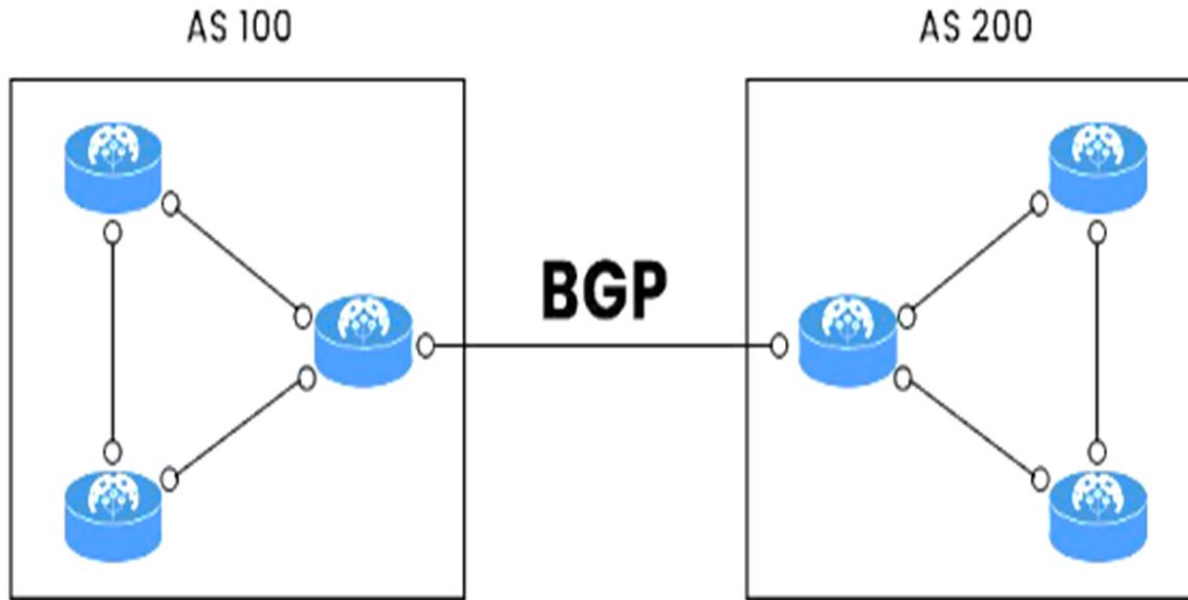
► Exterior Routing Protocol:

- ❖ Exterior gateway protocols are used to **exchange routing information** between autonomous systems and rely on **IGPs** to resolve routes within an autonomous system.
- ❖ Exterior Gateway Protocol (EGP) is a Routing Protocol which is used to find **network path information** between different networks.
- ❖ It is commonly used in the Internet to **exchange** routing table information between two **neighbour gateway hosts** (each with its own router) in a network of autonomous systems.
- ❖ **Border Gateway Protocol (BGP)** is the only Exterior Gateway Protocol (EGP) at the time.

► BGP(Broader Gateway Protocols):

- ❖ Border Gateway Protocol, is a **standardized exterior gateway protocol** used to exchange routing and reachability information **between different** autonomous systems (AS) on the internet.
- ❖ Also referred to as path vector protocol (**advanced distance vector**) protocols.
- ❖ IGP announce networks and describe the **cost to reach** those networks, BGP announces pathways and the networks that are reachable at the **end of the pathway**.
- ❖ BGP describes the pathway by **using attributes** which are similar to metrics.(ex. Next hop, origin etc.)
- ❖ TCP connection and **TCP port 179** are utilized by BGP in order to exchange data and messages.

► BGP(Broader Gateway Protocols):



► BGP(Broader Gateway Protocols):

❖ When BGP is not appropriate?

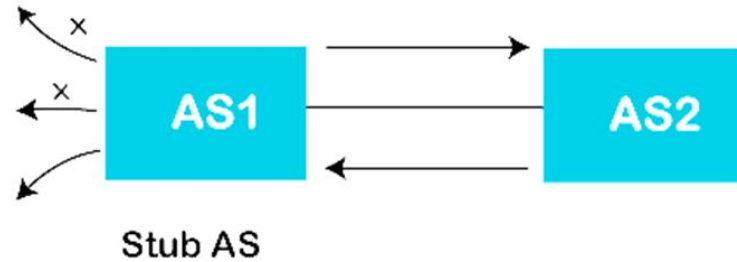
- 1-**Single connection** to Internet or other AS.
- 2-Lack of memory and processing power to handle updates.
- 3-Low bandwidth between ASs.
- 4-Limited understanding of route filtering & BGP path selection process.

❖ When BGP is most appropriate?

- 1-An AS allows packets to **transit through it** to reach other AS (e.g. Service Provider)
- 2-An AS has **multiple connections** to other AS
- 3-Routing policy & route selection for traffic entering or leaving the AS must be **manipulated**.

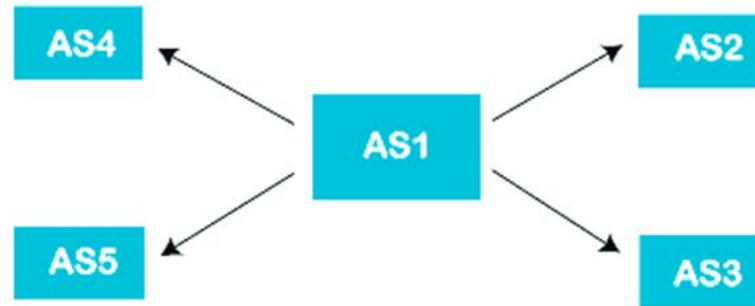
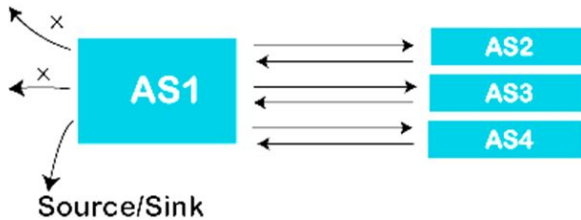
► Types of Autonomous System:

- ❖ Stub AS
- ❖ Multihomed AS
- ❖ Transient AS



◉ Transient Autonomous System

◉ Multihomed autonomous system



► BGP Tables:

❖ 1-Neighbor table:

- List of **BGP neighbours** "BGP peers" (configured statically with the neighbour command & can be **reachable**)

❖ 2-BGP forwarding database table:

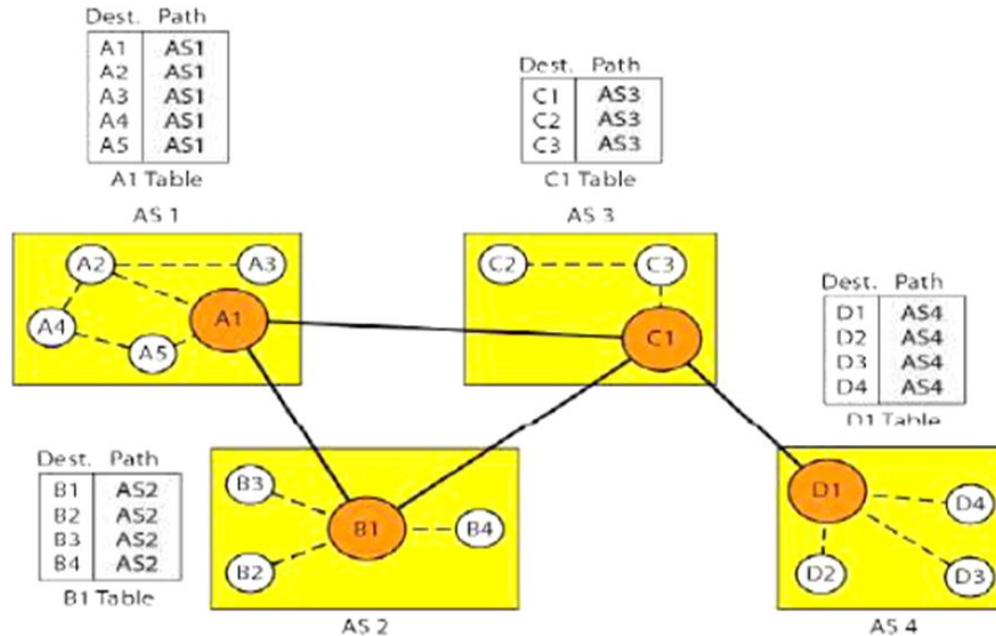
- List of all networks **learned** from each neighbour,
- Contain multiple paths to destination networks with attributes for each path
- Best paths in that table is **advertised to neighbours** in routing updates

❖ 3-IP routing table

- **List of best paths** to destination networks

► Initialization of Routing Table:

- ❖ Each speaker node have **information of other node** inside the AS only.



► Sharing & Updating of Routing Table:

❖ Sharing:

- Speaker nodes will share the information of **AS to other AS**.

❖ Updating:

- If a node (router or destination) receives multiple routes to reach it, the one with the **fewest number of intermediary Autonomous Systems** (ASes) in between is chosen.
- In simpler terms, the path with the **least number of stops** (ASes) is selected.

► Stable Routing Table:

Dest.	Path
A1	AS1
...	
A5	AS1
B1	AS1-AS2
...	...
B4	AS1-AS2
C1	AS1-AS3
...	...
C3	AS1-AS3
D1	AS1-AS2-AS4
...	...
D4	AS1-AS2-AS4

A1 Table

Dest.	Path
A1	AS2-AS1
...	
A5	AS2-AS1
B1	AS2
...	...
B4	AS2
C1	AS2-AS3
...	...
C3	AS2-AS3
D1	AS2-AS3-AS4
...	...
D4	AS2-AS3-AS4

B1 Table

Dest.	Path
A1	AS3-AS1
...	
A5	AS3-AS1
B1	AS3-AS2
...	...
B4	AS3-AS2
C1	AS3
...	...
C3	AS3
D1	AS3-AS4
...	...
D4	AS3-AS4

C1 Table

Dest.	Path
A1	AS4-AS3-AS1
...	
A5	AS4-AS3-AS1
B1	AS4-AS3-AS2
...	...
B4	AS4-AS3-AS2
C1	AS4-AS3
...	...
C3	AS4-AS3
D1	AS4
...	...
D4	AS4

D1 Table

► BGP path attributes:

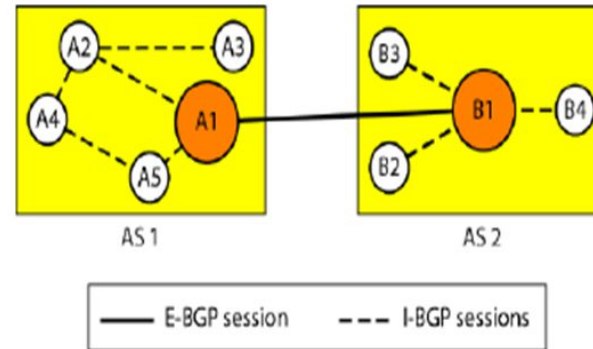
- ❖ Path attributes : List of **attributes used by BGP** to find best route.
- ❖ Well Known Attribute
 - Attributes that BGP router must recognize, Its categories are:
 - **Well known mandatory:** is one that **must appear** in the description of a route.
 - **Well known discretionary:** is one that **must be recognized** by each router, **but is not required** to be included in every update message.

► Optional Attributes:

- ❖ **Optional Attribute:** is one that needs **not be recognized** by every BGP router, Its categories are:
- ❖ **Optional transitive attribute:** is one that **must be passed** to the next router by the router that **has not implemented** this attribute.
- ❖ **Optional non-transitive attribute:** is one that **must be discarded** if the receiving router **has not implemented** it.

► BGP Sessions:

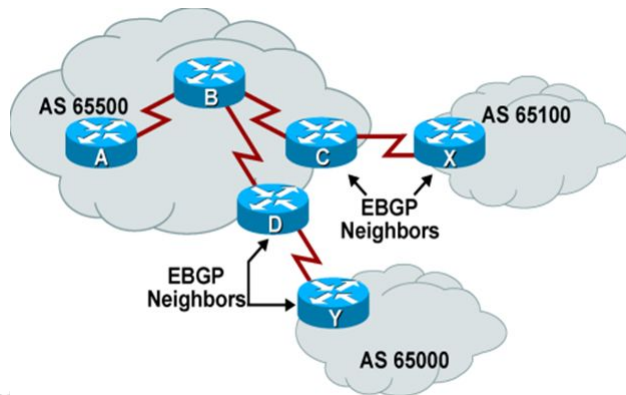
- ❖ A session is a **connection that is established** between two BGP routers only for the **sake of exchanging** routing information.
- ❖ BGP can have **two types** of sessions:
 - **1.E-BGP(External) session:** is used to exchange information between **two speaker nodes** belonging to **two different autonomous systems**
 - **2. I-BGP (Internal) session:** is used to exchange routing information **between two routers inside an autonomous.**



► BGP Sessions:

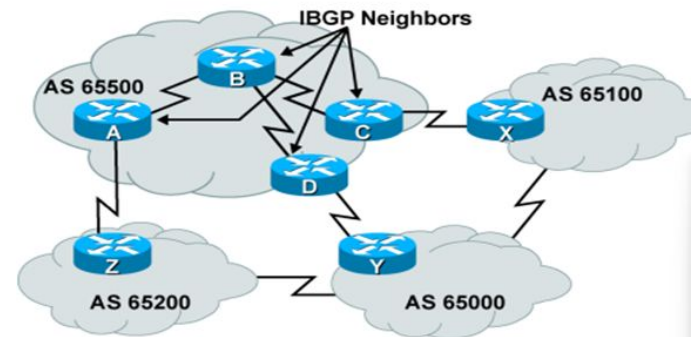
❖ External BGP

- When BGP neighbours belong to **different** autonomous systems they are called EBGP.
- E-BGP neighbours, by default, need to be **directly connected**.



❖ Internal BGP

- I-BGP refers to the presence of BGP neighbours within the **same AS**.
- The neighbours do not have to be directly connected, because they



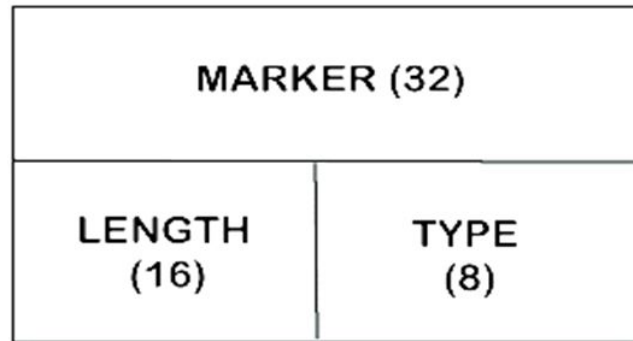
► BGP packets:

- ❖ **Open:** Starts session between routers like a hello packet.
- ❖ **Keep Alive:** Control if the neighbor is up or not.
- ❖ **Update:** Update packet is sent if **there is a change** in the network.
- ❖ **Notification:** This packet is sent if there is an **unwanted situation** and neighborhood is deleted.

► BGP Packets Formats:

- ❖ **Marker:** It is a 32-bit field which is **used for the authentication purpose.**
- ❖ **Length:** It is a 16-bit field that defines the **total length of the message**, including the header.
- ❖ **Type:** It is an 8-bit field that defines the **type of the packet.**

BGP Packet Format





Overview of IPv4 to ▶ IPv6 Transition Mechanisms

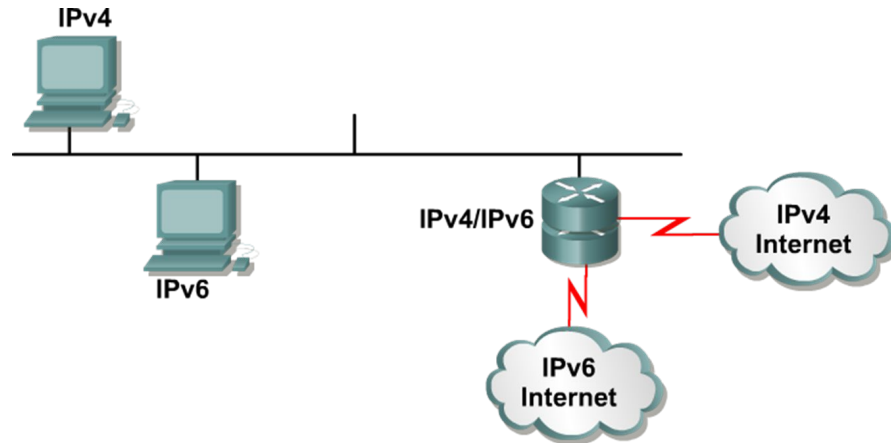
4.11

► Overview:

- ❖ In now a days, user of internets network are increasing like the bacteria production.
- ❖ A simple java enabled phone have the internet connectivity.
- ❖ There become a problem to identify the devices. So there is need of transition or migrate ipv4 to ipv6.
- ❖ The ipv6 is the (6th) **sixth version** of ip address.
- ❖ It is a 128 bit long value , allowing for 2^{128} addresses.
- ❖ To represent the 128 bit addresses,16 bit hexadecimal fields separated by colons.
- ❖ Techniques of migrating ipv4 to ipv6:
 - Dual Stack
 - Tunneling
 - Translation

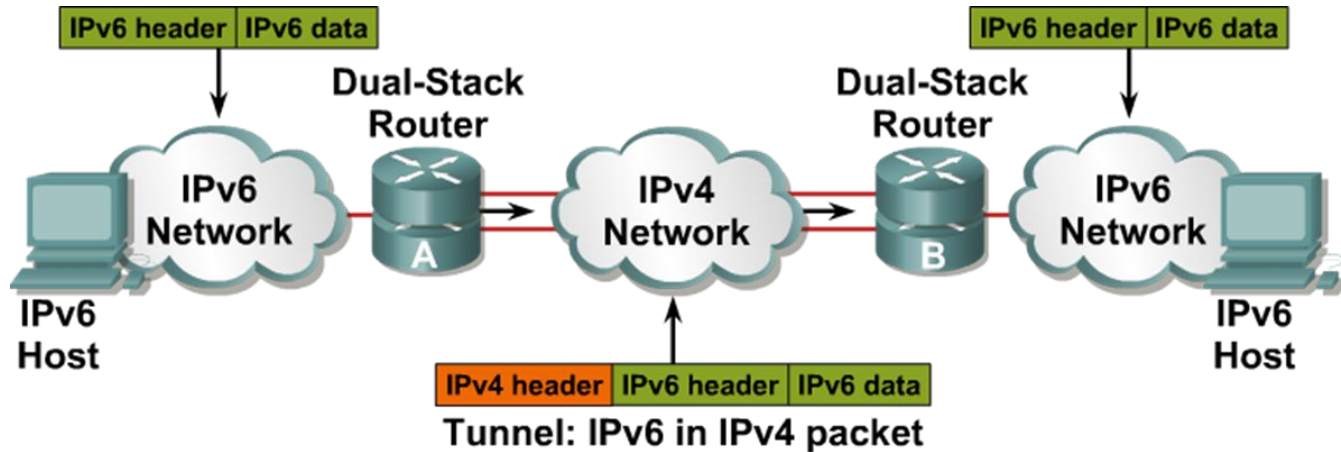
► Dual Stack:

- ❖ This allows all the **end hosts** and **intermediate network devices** (like routers, switches, modems etc.) to have **both IPv4 and IPv6** addresses and protocol stack.
- ❖ If **both the end stations support** IPv6, they can communicate using IPv6; otherwise they will communicate using IPv4.
- ❖ This will allow both IPv4 and IPv6 to **coexist and slow transition** from IPv4 to IPv6 can happen.



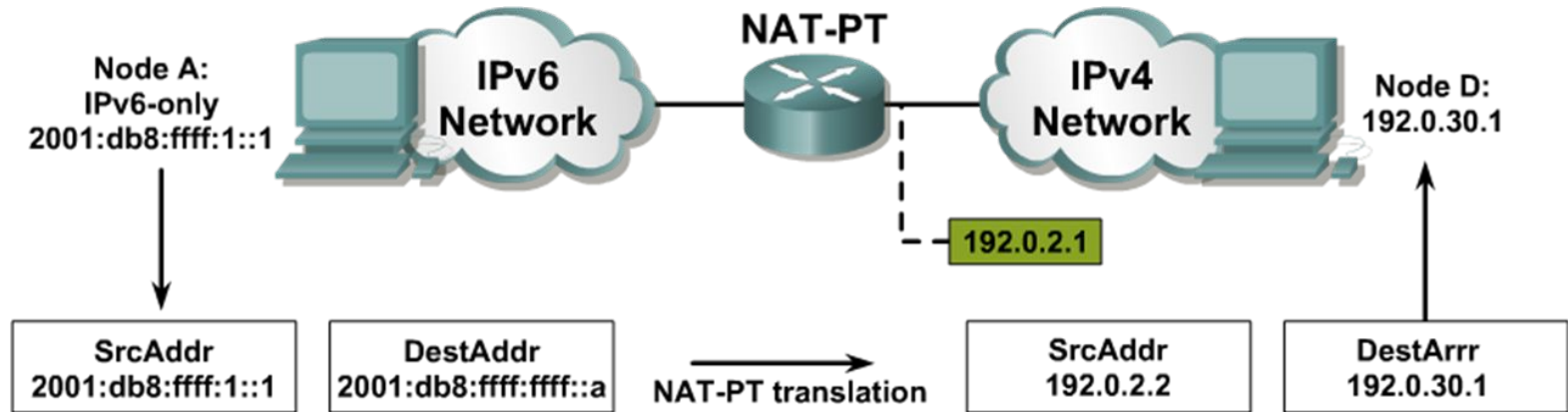
► Tunneling:

- ❖ This allows **encapsulating IPv6 packets in IPv4 packets** for transport over **IPv4 only** network.
- ❖ This will allow **IPv6 only** end stations to communicate over **IPv4 only** networks.



► Translation:

- ❖ This allows communication between **IPv4 only** and **IPv6 only** end stations.
- ❖ The job of the translator is to translate **IPv6 packets into IPv4 packets** by **doing address and port translation** and vice versa.

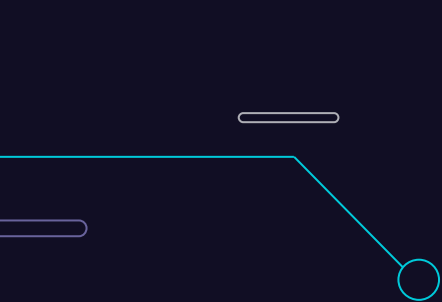


► Migration steps to Ipv6:

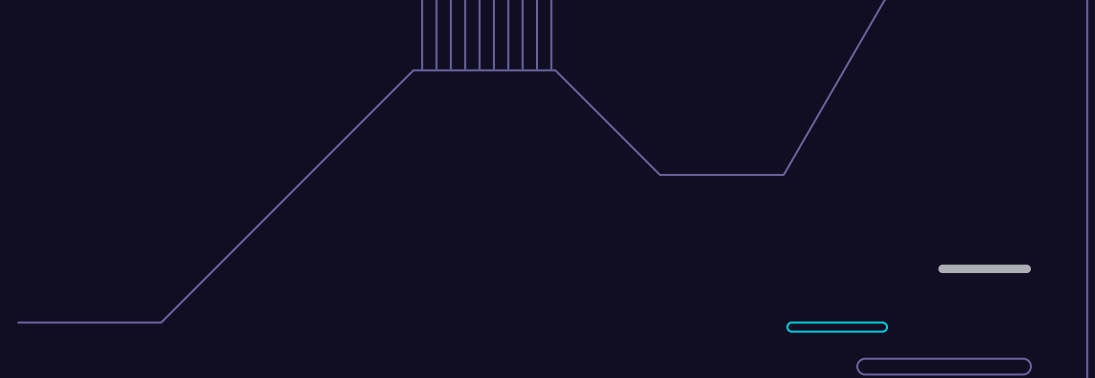
- ❖ There are **steps to migrate** ipv4 to ipv6.
 - Check IPv6 compliance
 - Plan IPv6 addressing
 - Enable IPv6 Routing
 - Setup IPv6 Application Servers
 - Enable IPv6 Peering
 - Migrate Services on IPv6

► BENEFITS OF MIGRATING TO IPv6:

- ❖ **Sufficient IP addresses:** IPv6 will create sufficient IP address pool for use by persons, as well as machines, facilitating the Internet of Things networks.
- ❖ **Efficient Routing:** due to reduced size of routing tables, elimination of the need for address translators in the network, and elimination of the need to perform error checking at various stages of data routes.
- ❖ **Transparency in Network:** due to the fact that each node in the network has a distinct address, which also makes troubleshooting easier.
- ❖ **Enhanced Security:** by the fact that IPv6 supports end to end IP Security protocol mode.
- ❖ **Efficient use of network bandwidth:** due to use of multicast as opposed to broadcast, when sending data to multiple destinations.



► Overview of ICMP/ICMPv6



4.1
2

► Overview of ICMP:

- ❖ **Internet Control Message Protocol** (both ICMPv4 and ICMPv6) is a protocol which acts as a **communication messenger protocol** between the communicating devices in IP network.
- ❖ ICMP messages provide **feedback**, **error reporting** and **network diagnostic** functions in IP networks which are necessary for the **smooth operation** of IPv6.
- ❖ **ICMPv6** is an integral part of IPv6 and ICMPv6 protocol in IPv6 has **much more** importance and functions than **ICMPv4** protocol in IPv4.
- ❖ **Main functions** of ICMPv6 are listed below:
 - Error Reporting
 - Network Diagnostics
 - Neighbor Discovery
 - Multicast Membership Reporting

► ICMP(Internet Control Messaging Protocols):

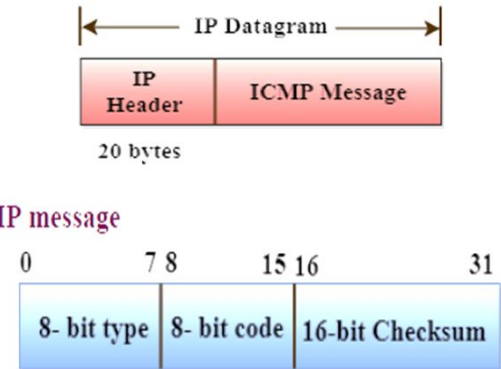
- ❖ The ICMP is a **network layer** protocol used by hosts and routers to send the **notifications** of IP datagram **problems** back to the sender.
- ❖ ICMP uses **echo test/reply** to check whether the destination is reachable and responding.
- ❖ ICMP handles both control and error messages, but its main function is to **report the error** but **not to correct** them.
- ❖ An IP datagram contains the addresses of both source and destination, but it does not know the address of the previous router through which it has been passed. Due to this reason, **ICMP can only send** the messages **to the source**, but not to the immediate routers.
- ❖ ICMP protocol communicates the error messages to the sender. ICMP messages cause the errors to be **returned back** to the user processes.
- ❖ ICMP messages are transmitted within IP datagram.

► ICMP(Internet Control Messaging Protocols):

❖ The first field specifies the **type of the message**.

- Type 0 – Echo/Reply
- Type 3 – Destination unreachable
- Type 5 – Redirect Message
- Type 8 – Echo Request
- Type 11 – Time Exceeded
- Type 12 – Parameter problem

The Format of an ICMP message

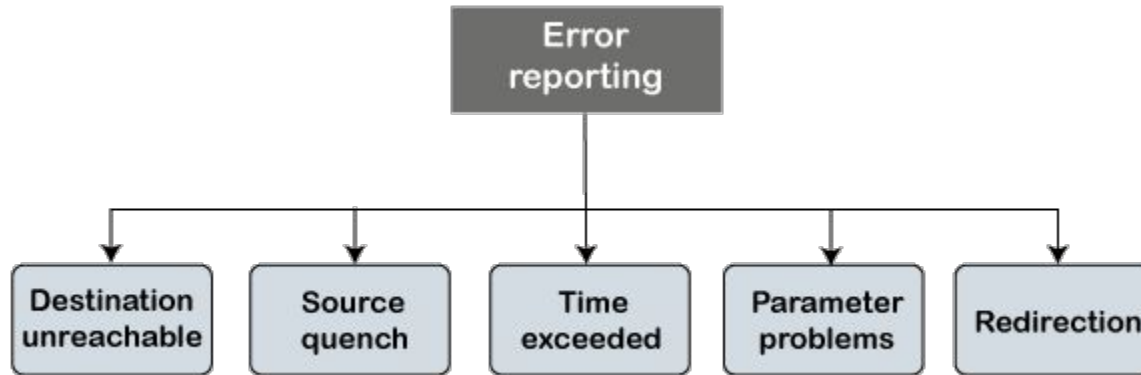


❖ The second field specifies the **reason** for a particular message type.

❖ The checksum field covers the **entire ICMP message**.

► ICMP(Internet Control Messaging Protocols):

- ❖ ICMP protocol reports the error messages to the sender.



► ICMP(Internet Control Messaging Protocols):

- ❖ **Destination unreachable:** The message of "Destination Unreachable" is sent from receiver to the sender when **destination cannot be reached**, or packet is discarded when the destination is not reachable.
- ❖ **Source Quench:** The purpose of the source quench message is **congestion control**. The message sent from the congested router to the source **host to reduce** the transmission rate. ICMP will take the IP of the discarded packet and then add the source quench message to the IP datagram to inform the source host to reduce its transmission rate. The source host will reduce the transmission rate so that the router will be **free from congestion**.
- ❖ **Time Exceeded:** Time Exceeded is also known as "**Time-To-Live**". It is a parameter that defines how long a packet should live before it would be discarded.

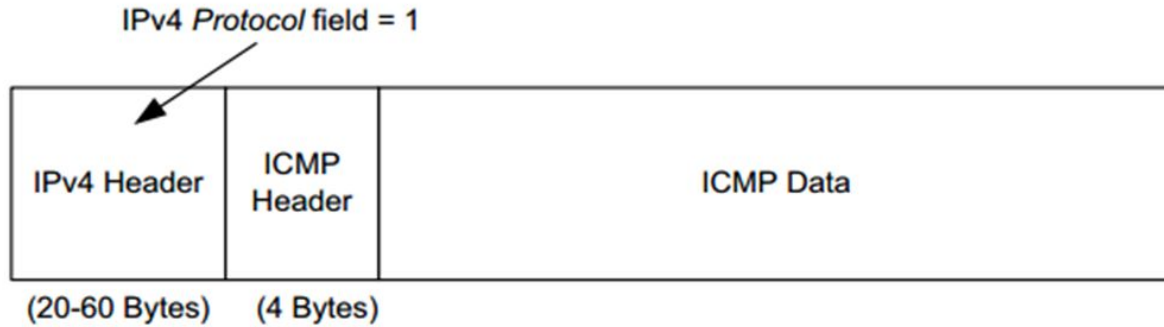
► ICMP(Internet Control Messaging Protocols):

- ❖ **Parameter problems:** When a router or host **discovers any missing value** in the IP datagram, the router discards the datagram, and the "parameter problem" message is **sent back** to the source host.
- ❖ **Redirection:** Redirection message is generated when host consists of a small routing table. When the host consists of a **limited number of entries** due to which it sends the datagram to a wrong router. The router that receives a datagram will forward a datagram to a correct router and also sends the "Redirection message" to the host to update its routing table.

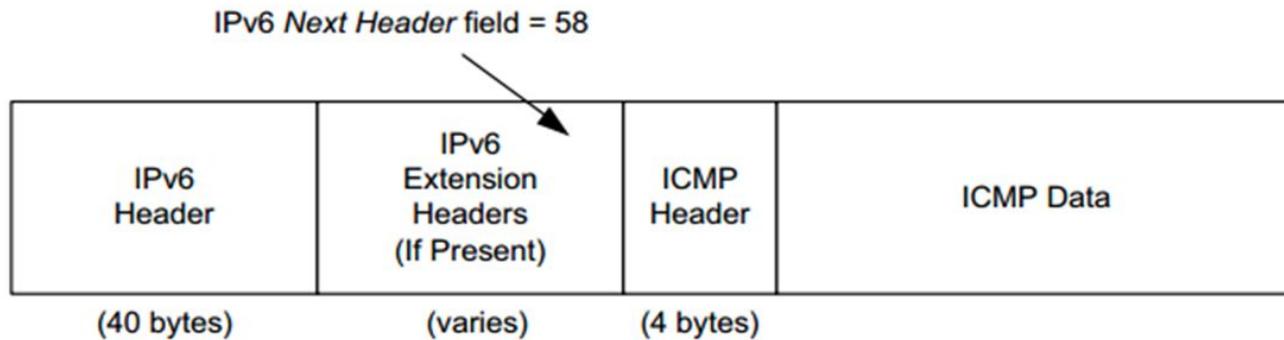
► ICMPv6:

- ❖ ICMPv6 removed the ICMPv4 functions that are **obsolete and took over** the functions of two major IPv4 protocols **IGMP** (Internet Group Membership Protocol) and **ARP** (Address Resolution Protocol).
- ❖ IGMP and ARP are **no more** with IPv6.
- ❖ ICMPv6, or Internet Control Message Protocol version 6, is a core protocol in the Internet Protocol suite used in IPv6 networks. ICMPv6 serves several functions, similar to its predecessor ICMP in IPv4, but it is designed specifically for IPv6.
- ❖ ICMPv6 messages are **encapsulated with IPv6 datagram packets** as shown below:

► ICMPv6:



IPv4



IPv6

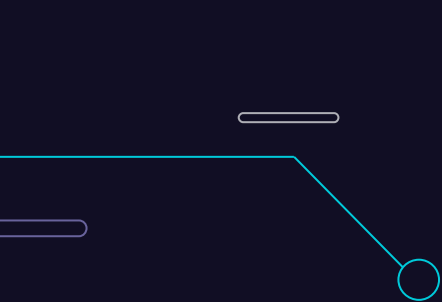
► Advantages / Disadvantages:

Advantages:

- ❖ Network devices use ICMP to **send error messages**, and administrators can use the Ping and Tracert commands to debug the network. These alerts are used by administrators to **identify issues** with network connectivity.
- ❖ **Network performance and connection monitoring** tools commonly employ ICMP to identify the existence of issues that the network team has to resolve.

Disadvantages:

- ❖ If the router drops a packet, it may be due to an error; but the way the IP (internet protocol) is designed, there is no way for the **sender to be notified** of this problem.



► Overview of Network Traffic Analysis



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► Overview of NTA:

- ❖ Network traffic analysis (NTA) involves the **monitoring and analysis** of data flowing over a computer network.
- ❖ Network Traffic Analysis (NTA) is the process of monitoring and analysing network data to **detect and respond** to security threats, performance issues, and other anomalies.
- ❖ The primary goal is to identify and respond to **abnormal patterns or behaviours** that may indicate security incidents, performance issues, or other network-related problems.
- ❖ **Tools** used for Network Traffic Analysis are:
 - Wireshark
 - Tcpdump
 - Snort
 - Darktrace

► TCPDump:

- ❖ **Type:** Open-source
- ❖ **Features:**
 - Packet-level analysis
 - Real-time capture and display
 - Support for a wide range of protocols
 - Protocol decoders for detailed analysis

► Wireshark:

- ❖ **Type:** Open-source
- ❖ **Features:**
 - Command-line packet analyzer
 - Real-time capture and display
 - Filtering options for targeted analysis



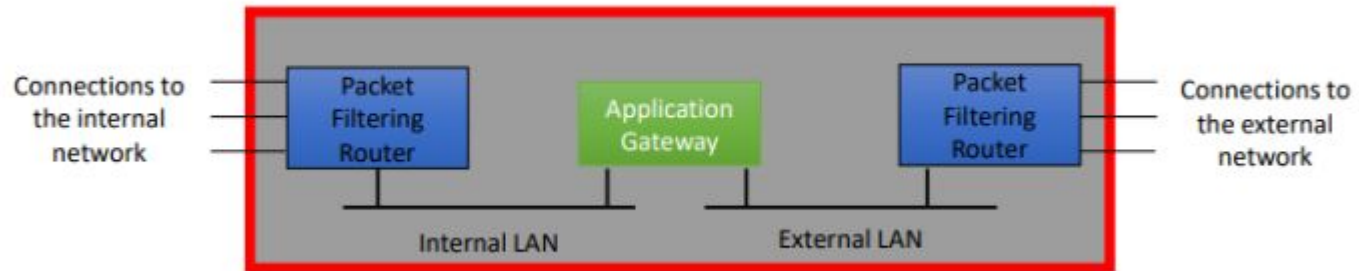
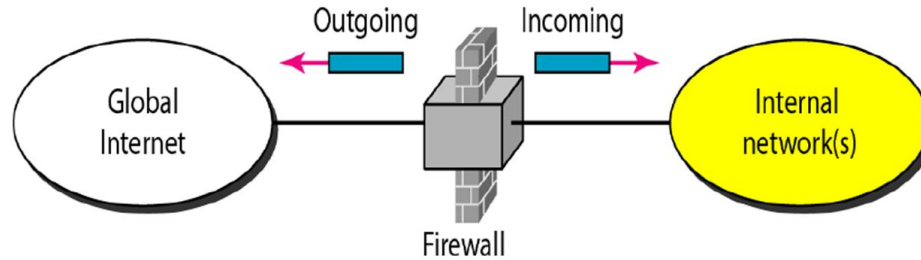
Security Concepts: ▶ Firewall & Router Access Control

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► Firewall:

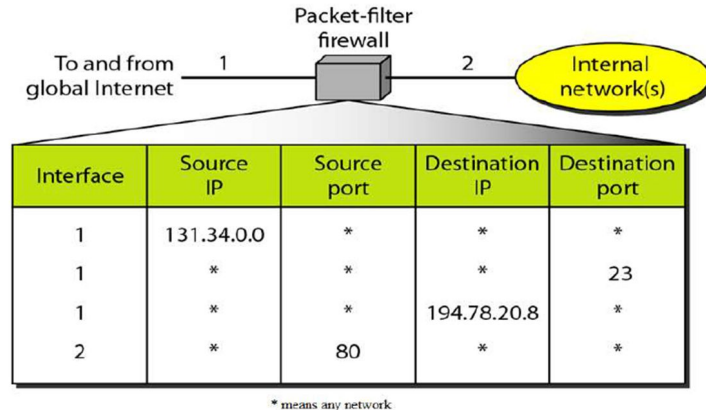
- ❖ A firewall is a **device** (usually a router or a computer) installed **between** the internal network of an organization and the rest of the Internet.
- ❖ Firewall is designed to **forward** some packets and **filter** (not forward) others .
- ❖ A firewall is a network security system designed to prevent **unauthorized access** to or from a private network
- ❖ Firewalls can be implemented in both **hardware and software**, or a combination of both.
- ❖ Network firewalls are frequently used to prevent unauthorized Internet users from accessing **private networks** connected.
- ❖ All messages entering or leaving the LAN **pass through** the firewall, which examines each message and blocks those that do not meet the specified **security criteria**.

► Firewall:



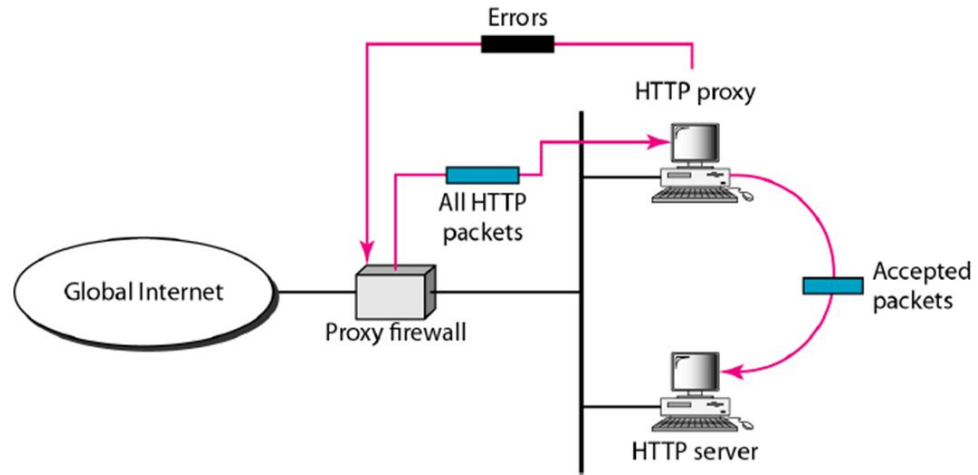
► Packet Firewall:

- ❖ A firewall can be used as a **packet filter**.
- ❖ It can forward or block **packets** based on the information in the network layer headers and transport layer **headers**: source and destination IP addresses, source and destination port addresses, and type of protocol(TCP /UDP).
- ❖ A packet-filter firewall is a router that uses a **filtering table** to decide which packets must be discarded(not forwarded).



► Proxy Firewall:

- ❖ Suppose, we need to filter a message based on the **information available** in the message itself (at the application layer).
- ❖ If **application level data** (like Text, URL, Websites, Links) is to be filtered then proxy firewall is best solution.



► Categories of Firewall:

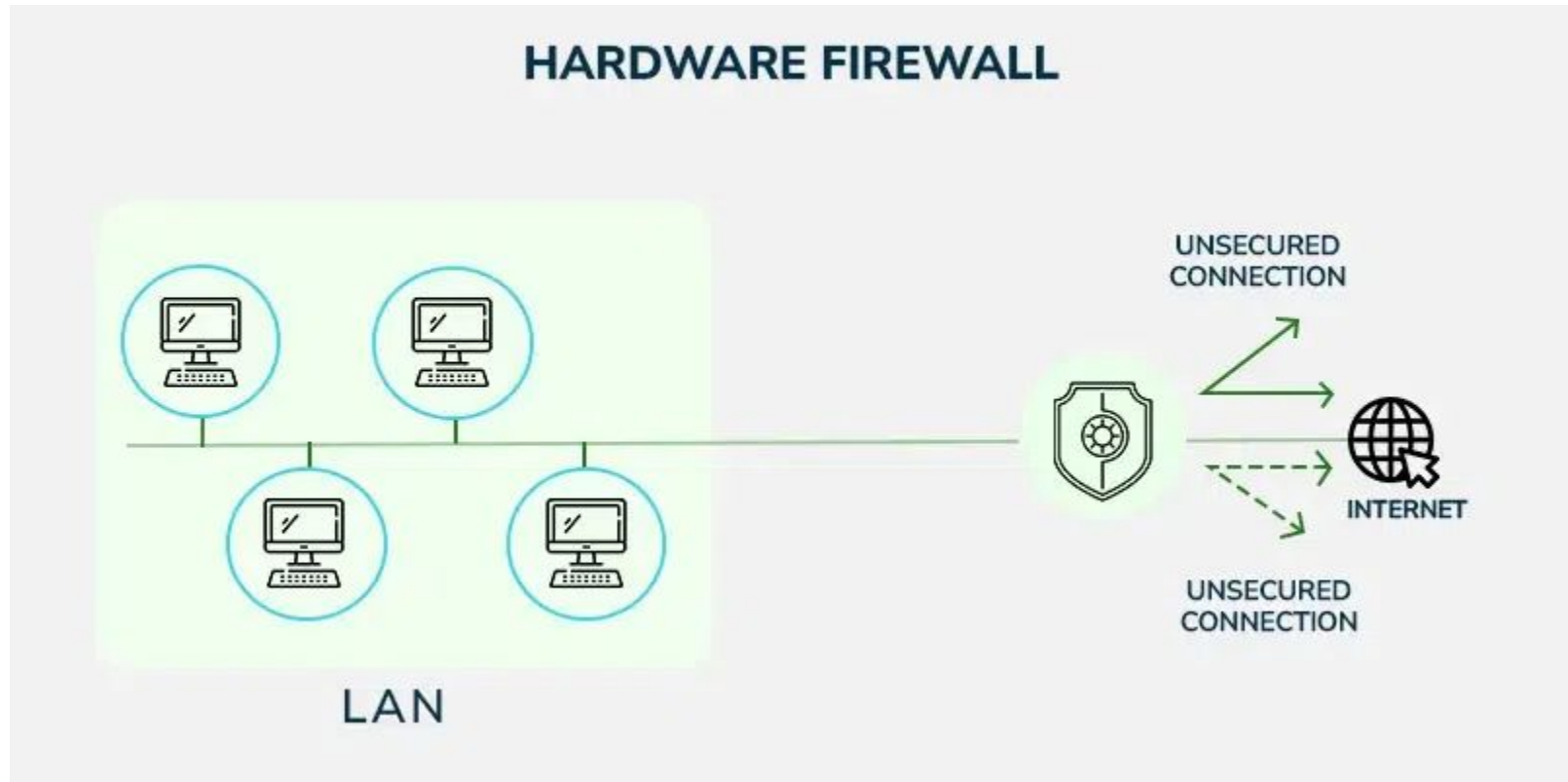
❖ Two **Categories of Firewall**

- Software firewall
- Hardware Firewall

❖ **Hardware Firewall**

- Hardware firewall is **dedicated device** which can be configured according to the need of large organization
- **Features:**
 - They are specialized devices
 - Hardware firewalls tend to be **expensive**.
 - Difficult to upgrade
 - Difficult to configure
 - **Example:** Cisco ASA, Palo Alto Network PA Series

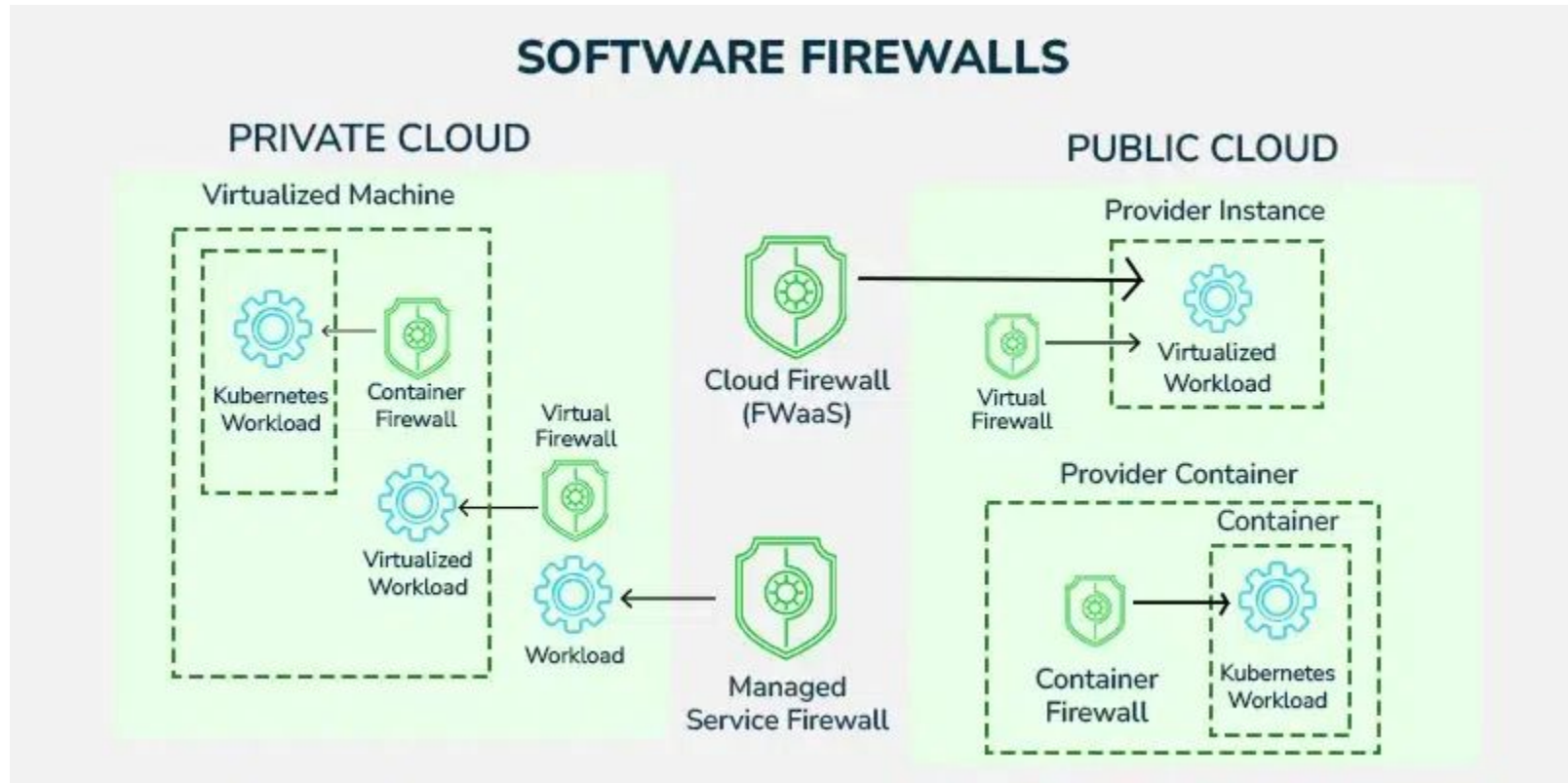
► Hardware Firewall:



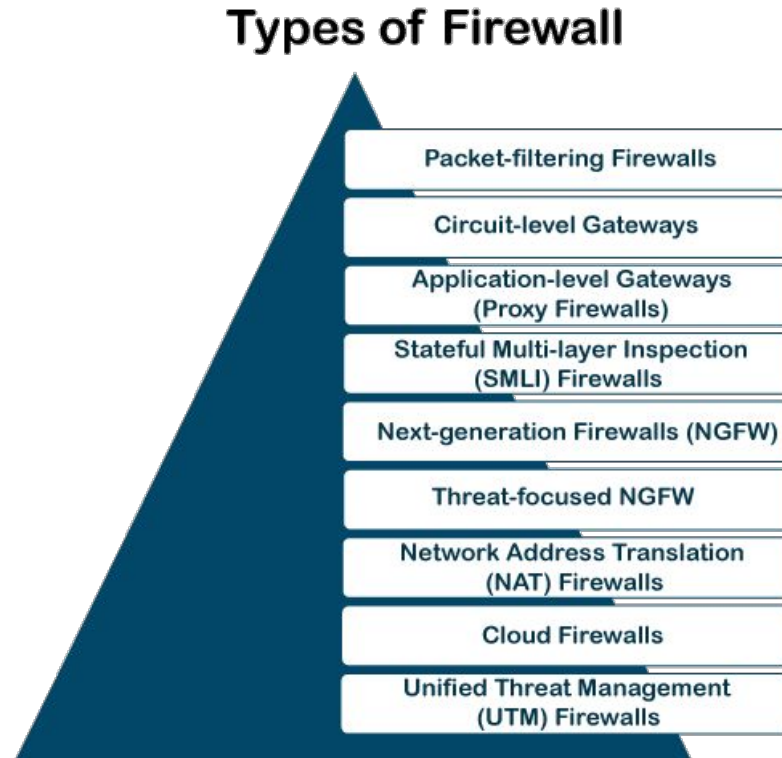
► Software Firewall:

- ❖ Software firewall can be **configured in PC**. It runs **as a program** in the PC and configuration can be done.
- ❖ **Features**
 - They are **not** specialized devices(Only special Programs in PC)
 - Less Complicated
 - Easy to upgrade
 - Easy to configure
 - Less expensive
 - **Examples:** Windows Defender Firewall, Norton 360, ZoneAlarm Etc

► Software Firewall:



► Types of Firewall:



► Firewall- Advantages/Disadvantages:

◆ Advantages of Firewalls

- **Security:** Firewalls block unauthorized access, protecting networks from hackers and malware.
- **Traffic Control:** They monitor and filter network traffic, ensuring only safe data passes through.
- **Data Protection:** Firewalls help prevent data breaches by limiting external access to sensitive information.

◆ Disadvantages of Firewalls

- **Performance Impact:** Filtering traffic can slow down network speed.
- **Complex Setup:** Configuring firewalls can be difficult, leading to potential security gaps.
- **Limited Internal Protection:** Firewalls are less effective against threats from inside the organization.

► PYQs:

- ❖ Provide reasons for transitions from IPv4 to IPv6. Describe the IPv6 datagram format. 2081
- ❖ Assume a class C network and divide it into three subnets. What is the value of the new subnet? 2081
- ❖ What is classful addressing? Explain link-state routing with a suitable example. 2080
- ❖ Short Notes on: 2080
 - IPV4
 - IPV6
- ❖ Difference between Unicast & Multicast routing. 2080
- ❖ Suppose you are given an IP address 192.168.0.0, perform subnetting and divide the given network in 2 subnets. Calculate total number of host that can be configured, range of IP address. 2080
- ❖ Why do you think network traffic analysis is carried out? How does IPv6 overcome the disadvantages of IPv4? 2080
- ❖ Is 192.16.144.64/27 a host, network, or broadcast address? In which layer of OSI model do Hub, Switch and Router operate on? 2080

► PYQs:

- ❖ Highlight on the importance of routing algorithm. Explain Distance Vector Routing algorithm and compare it with link state routing. 2080
- ❖ Differentiate between link state and distance vector routing. 2080
- ❖ Suppose you are given an IP address 172.16.0.0, perform subnetting and divide the given network into 2 subnets. Calculate total number of hosts that can be configured, range of the IP addresses. 2079
- ❖ What is classless addressing? Explain distance vector routing with a suitable example. 2078
- ❖ Write the subnet ID and broadcast address of each subnet if you divide a class C network (192.168.3.0 – 192.168.3.255) into 4 different subnets. What is the new subnet mask? 2078
- ❖ Explain distance vector routing with an example. 2079
- ❖ Explain subnetting and supernetting? Calculate all necessary parameters of subnetting a class C network (of your choice) into equal 4 subnets. 2078
- ❖ What is NAT? Explain it in detail and suggest how it can help in address depletion problem. 2078

► PYQs:

- ❖ What is routing table? Differentiate static routing table with dynamic routing table. 2078
- ❖ Explain the structure of IPv6 address. Compare IPv6 address with IPv4 address.
- ❖ What subnet ID and broadcast address of each subnet if you divide a class B network (150.10.0.0 – 150.10.255.255) in 4 different subnets. What is the new subnet mask? 2078
- ❖ Explain link state routing with example. 2078



▶ **THANKS!**

Do you have any questions?

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